

Unit 6

DEFORMATION OF SOLIDS

55 items: 22 from official papers (128 marks), 33 from the booklet

Syllabus sections in this unit

6.1 Stress and strain

6.2 Elastic and plastic behaviour

Unit 6: Deformation of solids

1. M/J 26/P22/Q2/c

- (b) A block rests in equilibrium on the centre of a uniform horizontal plank. The block is attached to two ropes, each of which passes over a fixed pulley and then attaches to an end of the plank as shown in Figure 2.1.

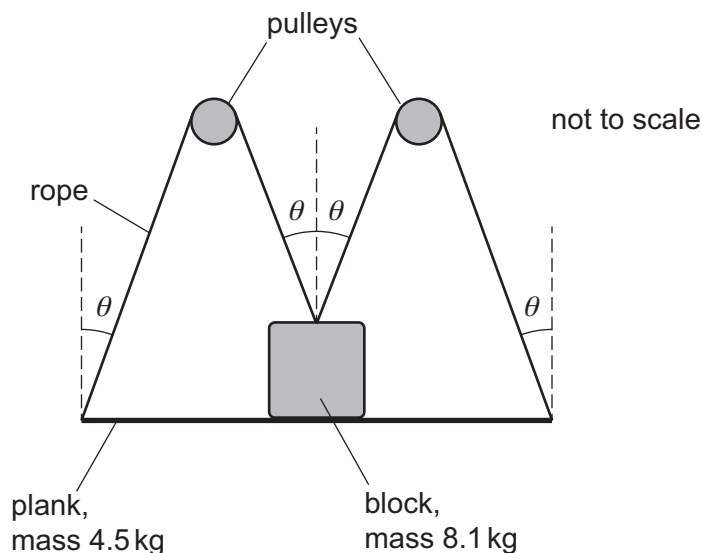


Figure 2.1

Where the ropes attach to the block and to the plank, the angle between the ropes and the vertical is θ . There is negligible friction between the pulleys and the ropes.

The mass of the block is 8.1 kg. The mass of the plank is 4.5 kg.

Figure 2.2 shows the directions of the forces acting on the plank and the block.

The weight of the plank is W_P , the weight of the block is W_B , the tension in each rope is T and the forces that the plank and the block exert on each other each have magnitude R .

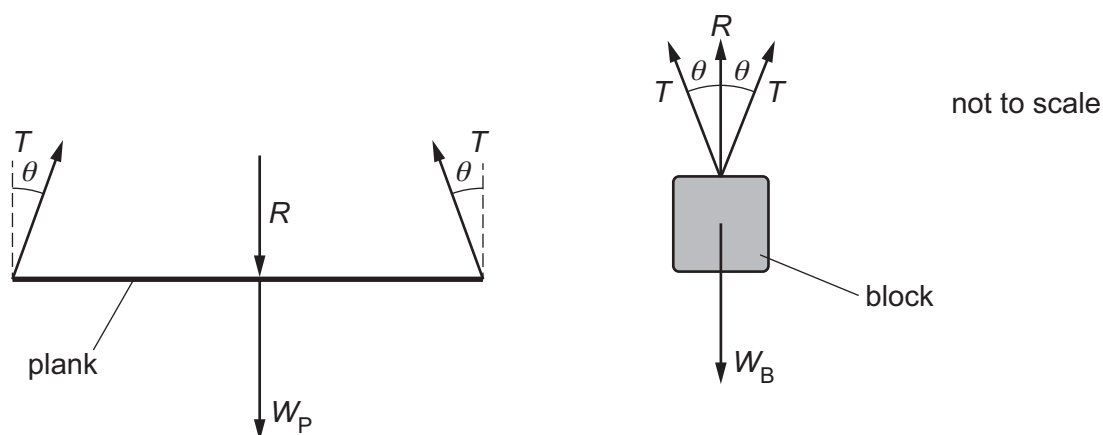


Figure 2.2

- (i) Use Figure 2.2 to show that

$$R + W_P = W_B - R.$$

[2]

- (ii) Use the equation in 2(b)(i) to calculate the magnitude of R .

$R =$

N [2]

(c) The extension of each rope in **2(b)** is proportional to the force applied to it.

(i) State the name of the law that describes this behaviour.

[1]

(ii) The ropes have elastic deformation.

State what is meant by elastic deformation.

[1]

2. M/J 26/P23/Q2/a,b(ii,iii),c,d

- 2 (a)** Define the Young modulus of a material.

[1]

- (b)** A long, thin thread of high-density polythene has a uniform cross-section. The thread is suspended from one end so that it hangs vertically.

The thread has a mass of 0.018 kg and a cross-sectional area of $2.6 \times 10^{-6} \text{ m}^2$.
High-density polythene has a density of 960 kg m^{-3} and a Young modulus of $8.0 \times 10^8 \text{ Pa}$.

- (ii)** Show that the stress at the top of the thread due to its own weight is $6.8 \times 10^4 \text{ Pa}$.

[2]

- (iii)** Figure 2.1 shows the variation with distance from the top of the thread of the stress in the thread due to its own weight.

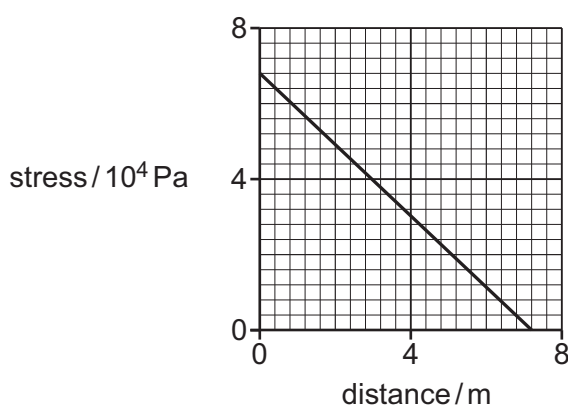


Figure 2.1

Suggest an explanation for the fact that the variation shown in Figure 2.1 is linear.

[2]

- (c)** The thread in **2(b)** is now laid horizontally on a frictionless bench. A tensile force is applied to the ends of the thread so that there is a uniform stress throughout the thread of $6.8 \times 10^4 \text{ Pa}$.

- (i)** Calculate the extension of the thread.

extension = _____ m [2]

- (ii)** Calculate the elastic potential energy of the thread.

elastic potential energy = _____ J [2]

- (d)** Suggest, with a reason, whether the total extension of the vertically suspended thread in **2(b)** is less than, equal to or greater than the answer in **2(c)(i)**.

[1]

3. M/J 26/P24/Q2/a,c**2 (a)** State Hooke's law.

[1]

- (b)** In an experiment to determine the spring constant of a spring X, a student measures the original length of the spring. The student then suspends an object in equilibrium from the spring and measures the final length of the spring.

The spring does not extend beyond its limit of proportionality.

The measurements taken by the student are shown in Table 2.1.

Table 2.1

original length of spring X	$(2.1 \pm 0.1) \text{ cm}$
final length of spring X	$(14.0 \pm 0.1) \text{ cm}$
mass of suspended object	$(0.300 \pm 0.005) \text{ kg}$

Calculate:

- (i)** the extension, in cm, of spring X together with its absolute uncertainty

extension = ($\pm$) cm [1]

- (ii)** the spring constant of spring X

spring constant = N m^{-1} [3]

- (iii)** the percentage uncertainty in the value of the spring constant calculated in **2(b)(ii)** due to the uncertainties in the measurements.

percentage uncertainty = % [2]

- (c)** An identical object to the one in **2(b)** is suspended in equilibrium from a spring Y that has a greater spring constant than spring X.

Spring Y does not extend beyond its limit of proportionality.

Describe and explain any difference in:

- (i)** the extension of the springs

[1]

- (ii)** the elastic potential energy of the springs.

[2]

4. MAR 26/P22/Q4/a(iv)

- 4** A uniform square picture is attached to a wire and hangs in equilibrium from a pin at point P, as shown in Figure 4.1.

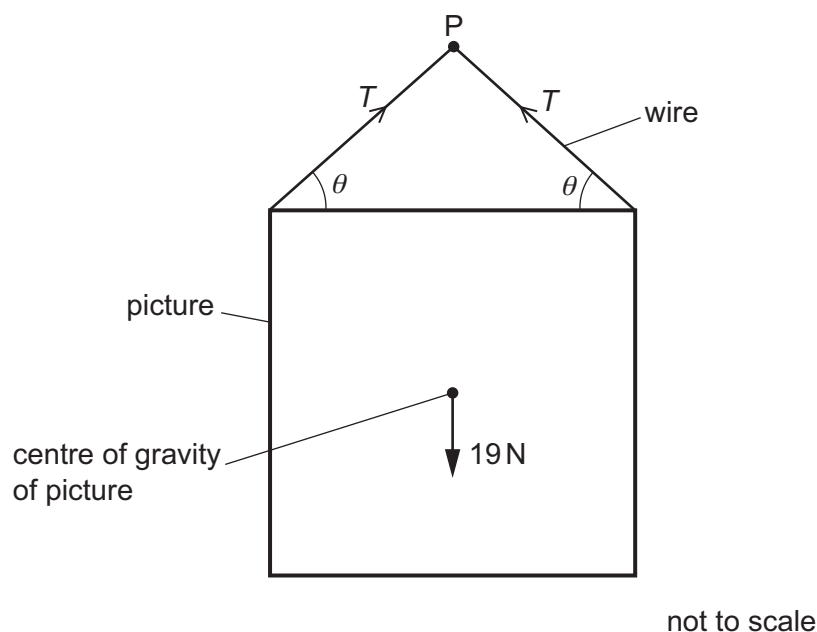


Figure 4.1

The picture has weight 19 N and its centre of gravity is directly below P.

The top side of the picture is horizontal. The wire forms an angle θ with the horizontal. The tension T in the wire is 13 N .

(a)

- (iv)** The wire has diameter 0.80 mm and is made of a metal with Young modulus $9.3 \times 10^{10}\text{ Pa}$. The wire obeys Hooke's Law.

Calculate the strain ε in the wire.

$\varepsilon =$

[3]

5. O/N 25/P21/Q3/a,b(ii),c(ii),d(ii)

- 3** A wire has length L and cross-sectional area A . The wire is made from a metal that has Young modulus E and resistivity ρ .

(a) Define the Young modulus of a material.

[1]

(b)

- (ii)** Show that the spring constant k_0 of the wire is given by

$$k_0 = \frac{EA}{L}.$$

[2]

- (c)** The wire is stretched, within the limit of proportionality, by a tensile force F . Assume that any changes in the cross-sectional area of the wire are negligible.

- (ii)** On Fig. 3.2, sketch the variation with F of the spring constant k of the wire.



Fig. 3.2

[1]

- (d)** Copper has a resistivity of $1.8 \times 10^{-8} \Omega \text{ m}$ and a Young modulus of $1.3 \times 10^{11} \text{ Pa}$.

A copper wire of diameter 1.6 mm has a resistance of 0.034Ω .

- (ii)** Use the equation in **(b)(ii)** to determine the spring constant of the wire.

spring constant =

N m^{-1} [2]

6. O/N 25/P22/Q3/b(ii),c

- 3** A spring is fixed at one end and attached to the frame of a pulley at the other end. A cable is passed around the wheel of the pulley. The spring is stretched to a fixed length using the cable and pulley.

Fig. 3.1 shows the view from above of the spring, cable and pulley.

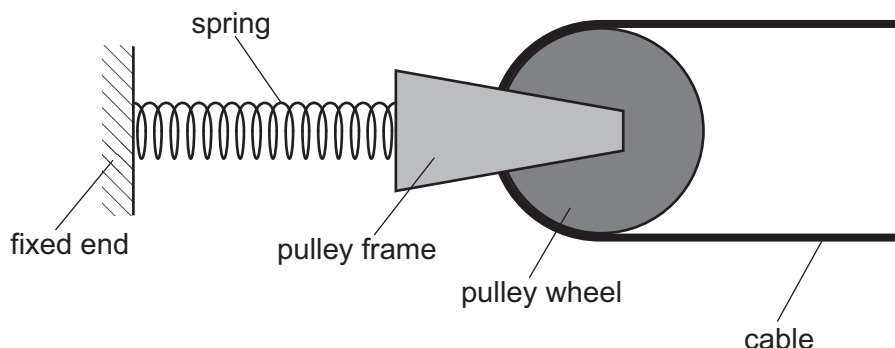


Fig. 3.1

The spring obeys Hooke's law and has a spring constant k of 250 N m^{-1} . A force F acts on the spring. The tension in the cable is T . The pulley is in equilibrium.

- (b)** The force F is 110 N .

(ii) Calculate the extension of the spring.

extension = _____ m [2]

- (c)** A second identical spring with the same spring constant of 250 N m^{-1} is now also connected to the pulley, as shown in Fig. 3.3.

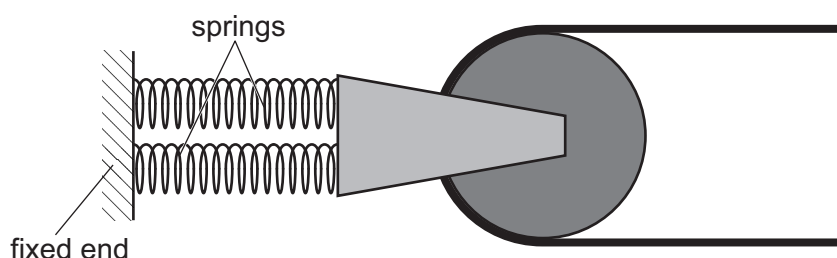


Fig. 3.3

The tension in the cable is kept the same. The pulley is again in equilibrium.

- (i)** Determine the extension of the springs.

extension = _____ m [2]

- (ii)** The elastic potential energy stored in the spring in Fig. 3.1 is E_1 . The **total** elastic potential energy stored in the two springs in Fig. 3.3 is E_2 .

Calculate the ratio $\frac{E_1}{E_2}$.

ratio = _____ [2]

7. O/N 25/P24/Q3/a,b,c(ii)

- 3 A bungee jumper of mass 64 kg secures one end of an elastic rope to a bridge. The other end is attached to the bungee jumper. The jumper falls from rest from the bridge and descends into the valley below, as shown in Fig. 3.1.

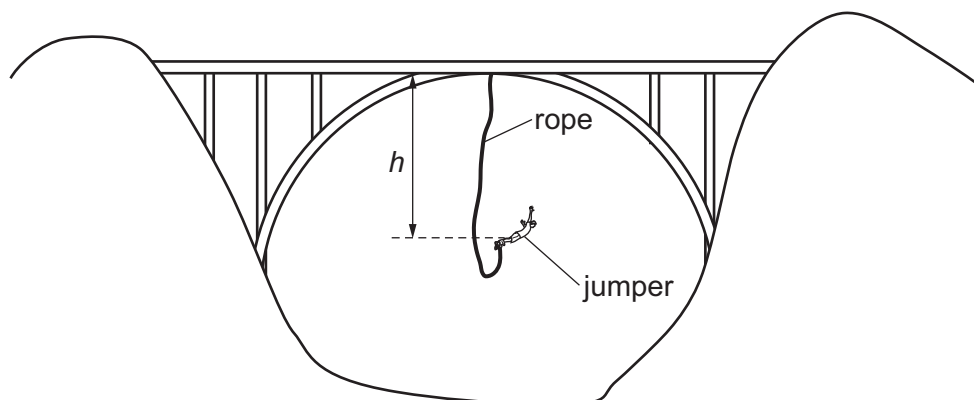
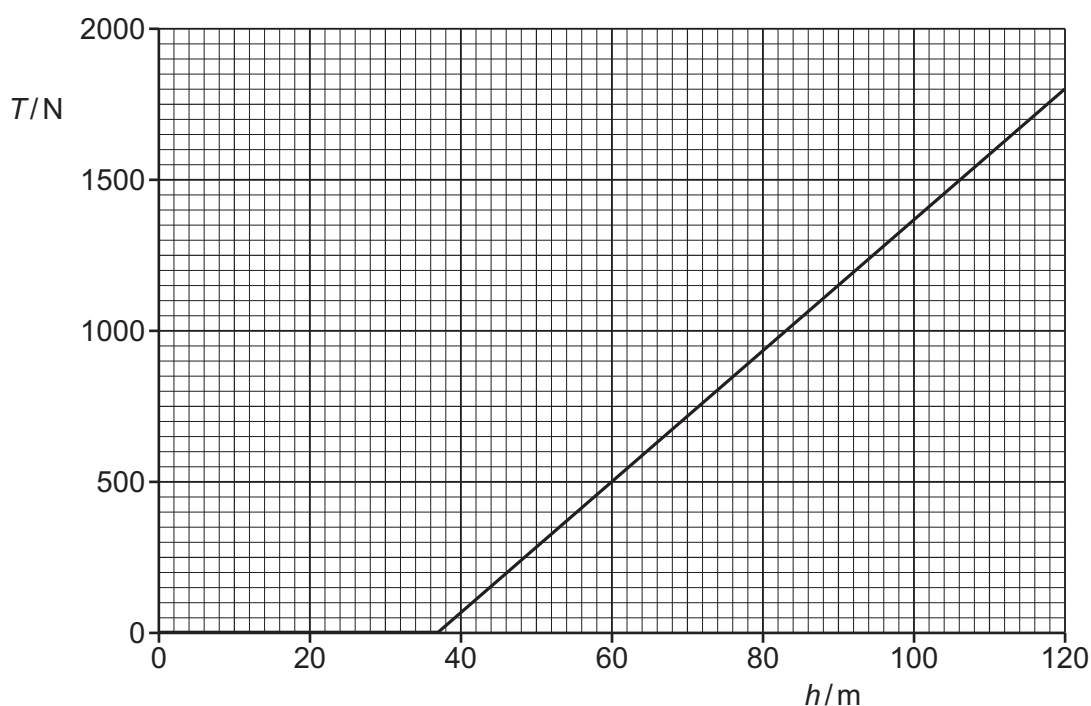
**Fig. 3.1** (not to scale)

Fig. 3.2 shows the variation of the tension T in the rope with the vertical distance h of the jumper below the level of the bridge.

**Fig. 3.2**

- (a) The rope obeys Hooke's law.

State Hooke's law.

[1]

- (b) (i) Determine the unstretched length of the rope.

length =

m [1]

- (ii) Determine the spring constant k of the rope.

$k =$

Nm^{-1} [2]

- (c) For the position of the bungee jumper at a distance of 120 m below the bridge:

- (ii) show that the elastic potential energy in the rope is 75 kJ.

[2]

8. M/J 25/P21/Q2/c

- (c) The beam is 0.80 m above horizontal ground.

Object A is removed and replaced by a spring connected to the ground and the beam, as shown in Fig. 2.2.

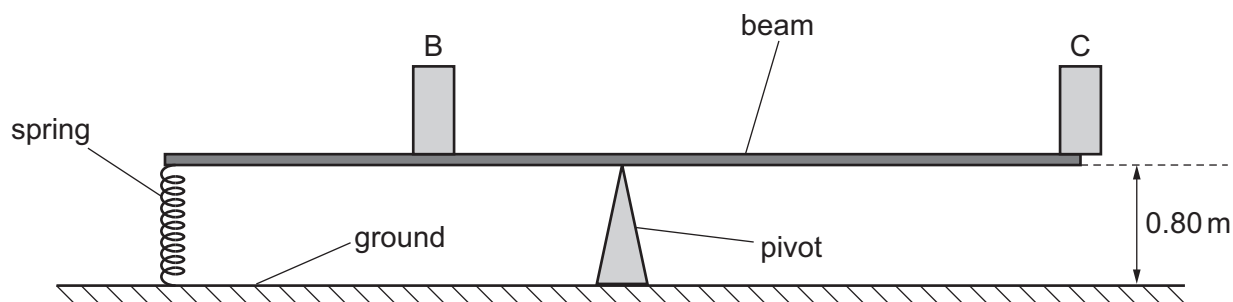


Fig. 2.2

After the change, the beam is again horizontal and in equilibrium. The positions of B and C are unchanged.

The spring has an unstretched length of 0.59 m and obeys Hooke's law.

- (i) Calculate the spring constant of the spring.

spring constant = Nm^{-1} [3]

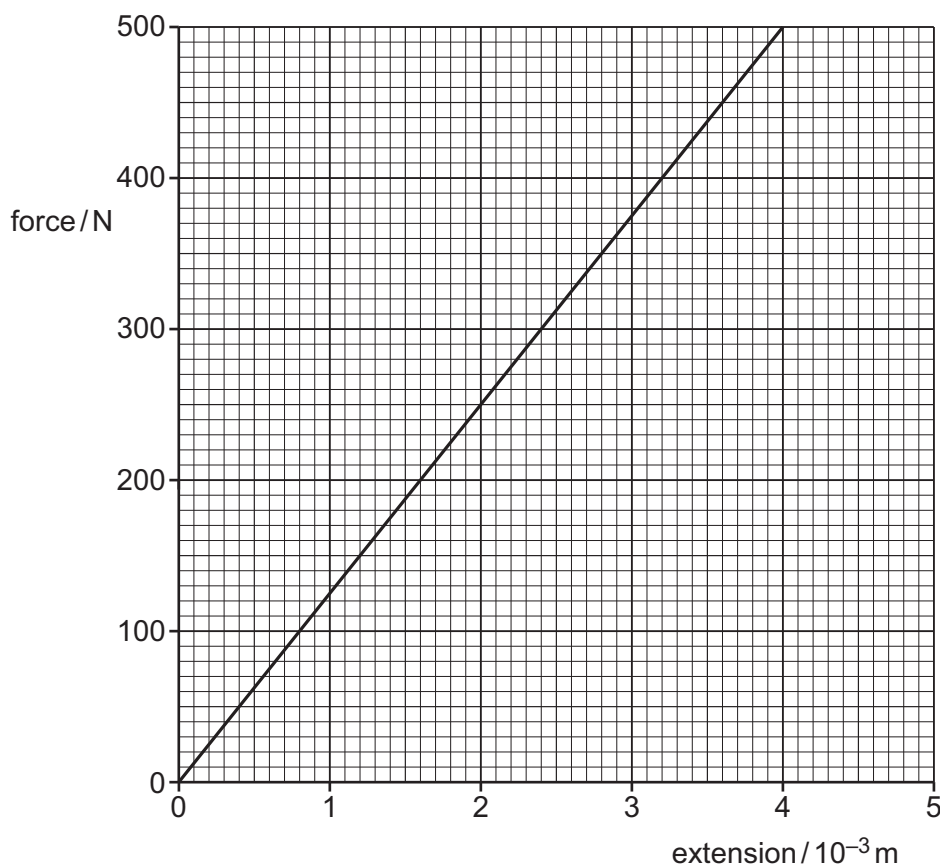
- (ii) Calculate the elastic potential energy of the spring.

elastic potential energy = J [2]

9. M/J 25/P22/Q5/a,b**5 (a)** Define the Young modulus.

[1]

(b) A wire of unstretched length 0.81 m is made of a metal with Young modulus 95 GPa. The wire obeys Hooke's law and has a constant cross-sectional area. Fig. 5.1 shows the force–extension graph for the wire.

**Fig. 5.1**

(i) Determine the cross-sectional area of the wire.

area = m² [3]

(ii) The extension of the wire is initially 2.0×10^{-3} m.

Determine the work done to increase the extension of the wire to 3.0×10^{-3} m.

work done = J [3]

10. M/J 25/P23/Q2/b(ii,iii)

- (b) Three objects A, B and C are placed on a horizontal beam. The beam is in equilibrium, as shown in Fig. 2.1.

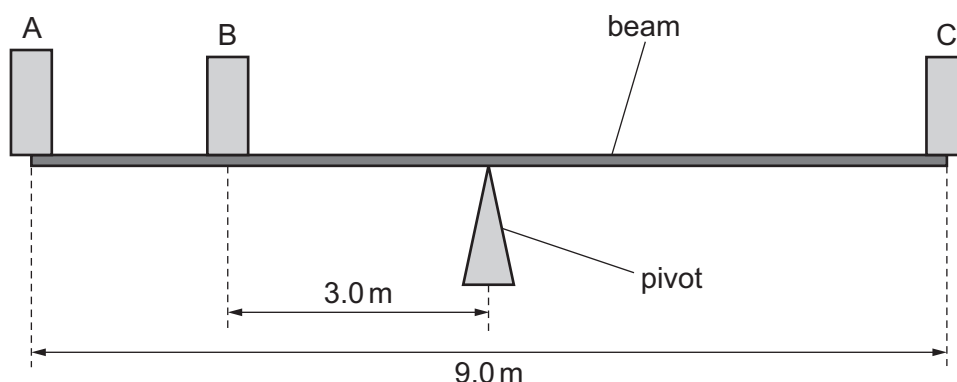


Fig. 2.1 (not to scale)

The beam is uniform and has length 9.0 m.

A pivot is at the midpoint of the beam.

Object A has mass 90 kg and is at one end of the beam.

Object B has mass m and is a distance of 3.0 m from the pivot.

Object C has mass 150 kg and is at the other end of the beam.

- (ii) Object A is removed and replaced by a wire fixed to the end of the beam and to the ground, as shown in Fig. 2.2.

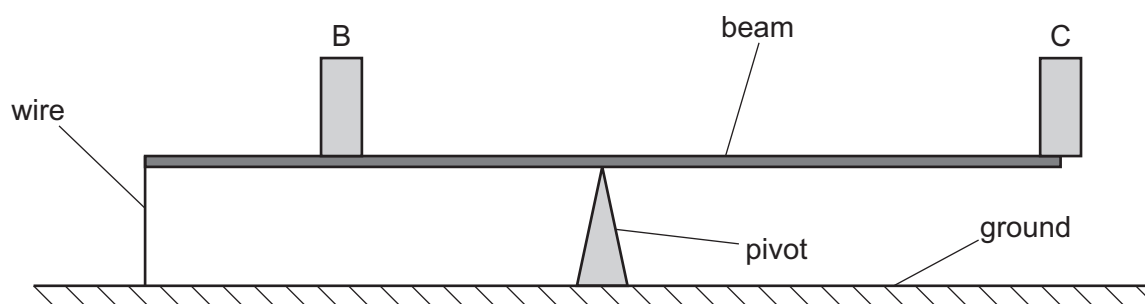


Fig. 2.2

After the change, the beam is again horizontal and in equilibrium. The positions of B and C are unchanged.

The wire has a diameter of 1.8×10^{-3} m and has a strain of 1.2×10^{-3} .

The wire is not extended beyond its limit of proportionality.

Calculate the Young modulus of the wire.

Young modulus = Pa [3]

- (iii) Object B is now moved to a new position closer to the pivot without passing it. The beam is again horizontal and in equilibrium.

State and explain the effect, if any, that this has on the strain in the wire.

[2]

11. M/J 25/P24/Q3/d

- 3** The lower end of a vertical spring is fixed to a horizontal surface, as shown in Fig. 3.1.

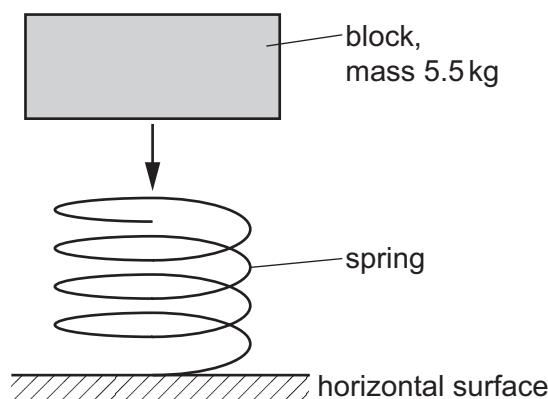


Fig. 3.1

The mass of the spring is negligible. A block of mass 5.5 kg drops vertically onto the spring and is brought to rest as the spring is compressed.

- (a)** The block has kinetic energy 110 J as it makes contact with the spring.

Calculate the speed of the block as it makes contact with the spring.

$$\text{speed} = \quad \text{ms}^{-1} \quad [2]$$

- (b)** The gravitational potential energy of the block decreases by 20 J as the spring is compressed to its maximum compression x_0 .

Show that x_0 is 0.37 m.

[2]

- (c)** Assume that, as the spring compresses, all of the energy lost by the block is converted into the elastic potential energy of the spring.

Use the data from **(a)** and **(b)** to determine the maximum elastic potential energy of the spring.

Show your working.

$$\text{maximum elastic potential energy} = \quad \text{J} \quad [1]$$

- (d)** The variation of the force F acting on the spring with the compression x of the spring is shown in Fig. 3.2.

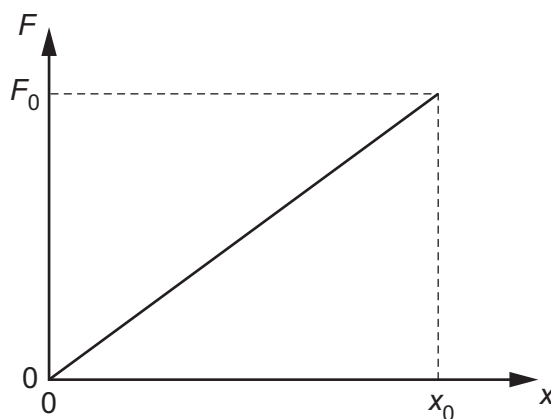


Fig. 3.2

Use the information in **(b)** and your answer in **(c)** to show that the maximum force F_0 exerted on the spring by the block is 700 N.

[2]

12. O/N 24/P21/Q4/a,b,c

- 4 (a) Define the Young modulus of a material.

[1]

(b) A metal wire P that obeys Hooke's law is stretched within its limit of proportionality.

- (i) On Fig. 4.1, sketch the variation of tensile force F in the wire with its extension x .

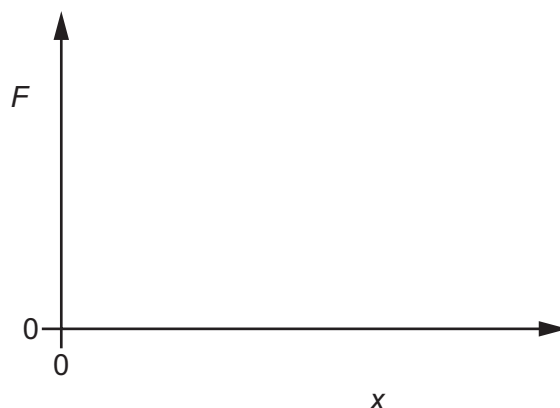


Fig. 4.1

[1]

- (ii) State the name of the quantity represented by the gradient of the line in Fig. 4.1.

[1]

- (iii) State the name of the quantity represented by the area under the line in Fig. 4.1.

[1]

- (c) Another wire Q is made from a metal that has twice the Young modulus of the metal of wire P in **(b)**. Wire Q has the same volume as wire P but has double the cross-sectional area of wire P.

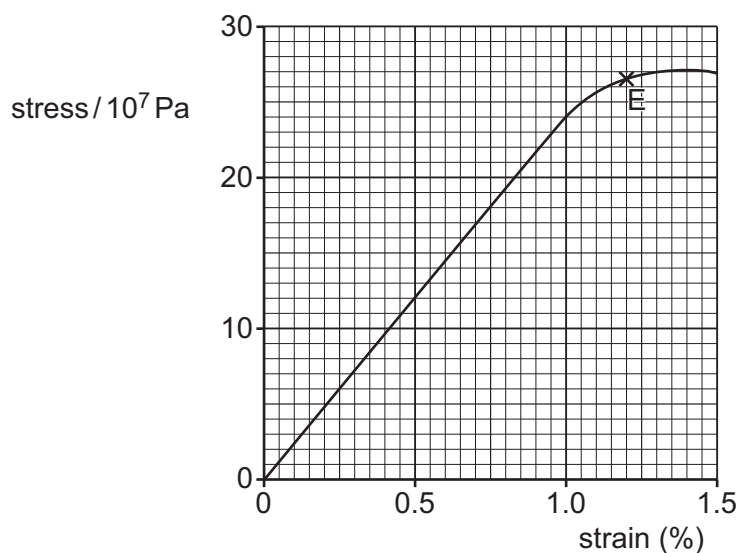
The two wires are extended by equal tensile forces within their limits of proportionality.

State and explain how the extension of wire Q compares with the extension of wire P.

[3]

13. O/N 24/P22/Q3/a,c(ii,iii)

- 3 (a) The variation of stress with strain for a metal P is shown in Fig. 3.1.

**Fig. 3.1**

Point E is the elastic limit of the metal.

- (i) Use Fig. 3.1 to determine the Young modulus for P.

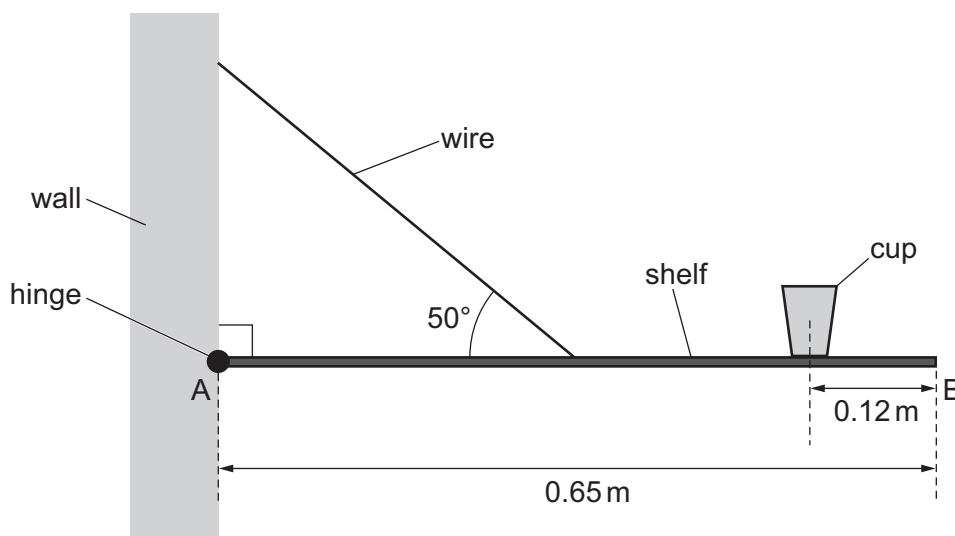
Young modulus =

Pa [2]

- (ii) On the line in Fig. 3.1, draw a cross (×) to show the limit of proportionality. Label this point Q.

[1]

- (c) A wire is used to hold a uniform shelf AB horizontally in equilibrium as shown in Fig. 3.2.

**Fig. 3.2** (not to scale)

The wire is connected to the midpoint of shelf AB at an angle of 50° to the horizontal. The shelf is attached to a wall by a hinge at A. The length of shelf AB is 0.65 m and its weight is 33 N.

A cup of weight 1.5 N rests on the shelf with its centre of gravity at a horizontal distance of 0.12 m from B.

- (ii) The stress in the wire is 1.5×10^7 Pa.

Determine the radius of the wire.

radius =

m [2]

- (iii) More items are added to the shelf, doubling the stress in the wire. The wire is made of the metal P from (a).

Use Fig. 3.1 to state and explain whether the wire will behave plastically or elastically as the stress doubles.

[2]

14. O/N 24/P23/Q4/a,b

4 (a) Define:

(i) stress

[1]

(ii) strain.

[1]

- (b) Two wires X and Y, with equal unstretched lengths of 0.84 m, are suspended from fixed points that are at the same horizontal level. The lower ends of the wires are attached to a beam of negligible mass. The beam is horizontal and in equilibrium, as shown in Fig. 4.1.

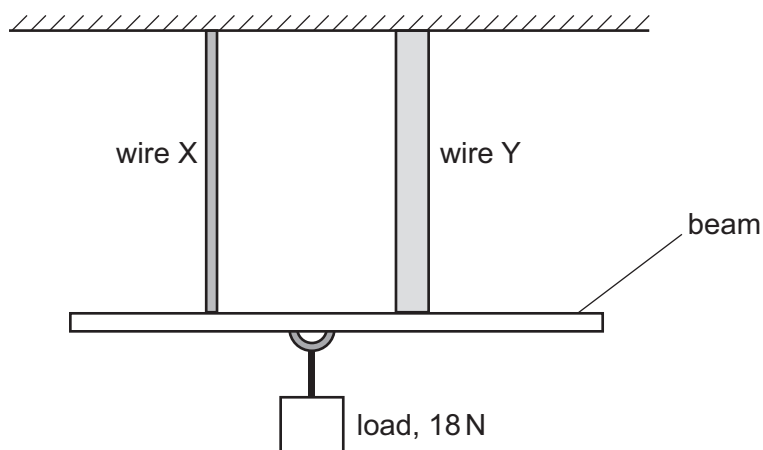


Fig. 4.1

Wire X is made from a metal that has a Young modulus of 1.9×10^9 Pa.
Wire Y is made from a different metal.

A load of weight 18 N is suspended from the beam at a point that is equidistant from the two wires. This load causes both wires to extend by 0.47 mm.

- (i) Determine the cross-sectional area of wire X.

cross-sectional area = m² [3]

- (ii) Wire Y has a greater diameter than wire X.

Explain, without calculation, whether the Young modulus of the metal from which wire Y is made is less than, the same as or greater than 1.9×10^9 Pa.

[2]

15. M/J 24/P21/Q4/a,b,c

4 (a) Define strain.

[1]

(b) A copper wire of length 4.0 m has a uniform cross-sectional area of $4.5 \times 10^{-7} \text{ m}^2$.

A tensile force of 18 N is applied to the wire. This causes the wire to extend by 1.4 mm up to its limit of proportionality.

(i) Calculate the Young modulus of the wire.

Young modulus = Pa [3]

(ii) On Fig. 4.1, draw a line to show how the stress varies with the strain for the wire up to its limit of proportionality.

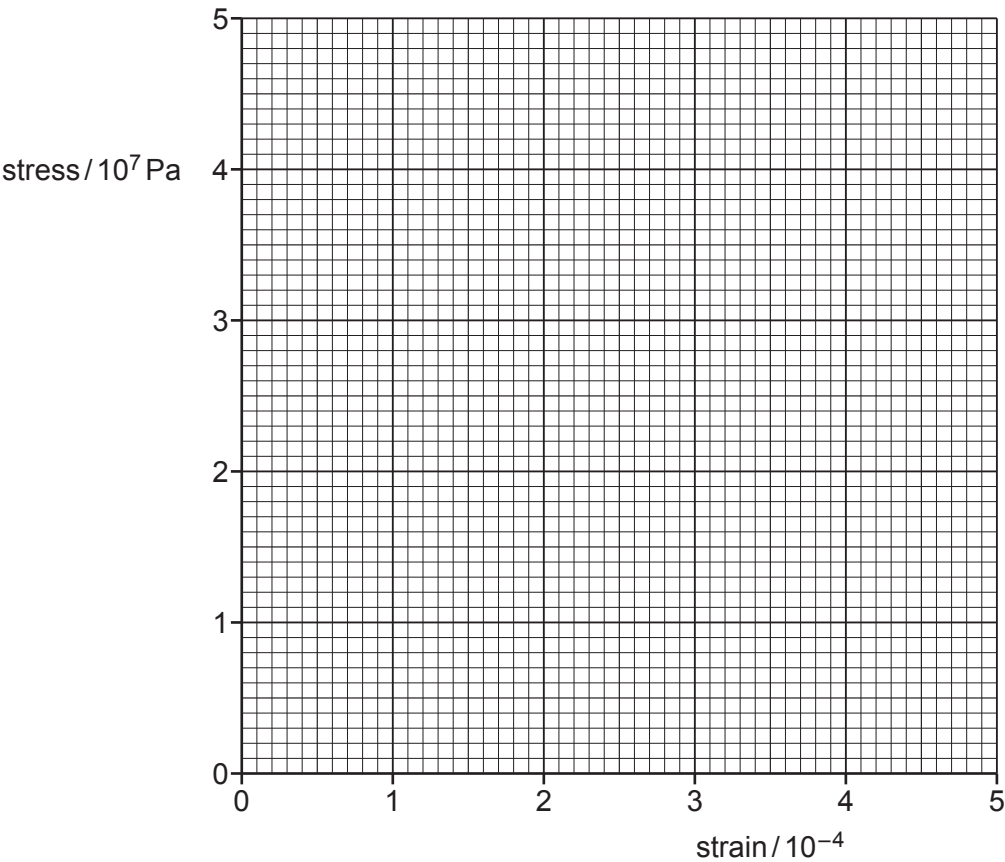


Fig. 4.1

[2]

(c) A second copper wire has the same length as the wire in (b) but a larger diameter. Both wires are subjected to a tensile force of 18 N.

By placing a tick (✓) in each row, complete Table 4.1 to compare the stress and strain of the two wires.

Table 4.1

	greater in second wire	less in second wire	the same in both wires
stress			
strain			

[2]

16. M/J 24/P22/Q3/d

- 3 Lightning occurs when charge builds up in the atmosphere, creating a potential difference between the ground and the atmosphere.

During a lightning strike there is an average current of $3.3 \times 10^4 \text{ A}$ for a time of $2.6 \times 10^{-5} \text{ s}$.

- (d) A section of the lightning rod of length 0.12 m is removed for testing. A tensile stress of $1.9 \times 10^6 \text{ Pa}$ is applied, as shown in Fig. 3.2.

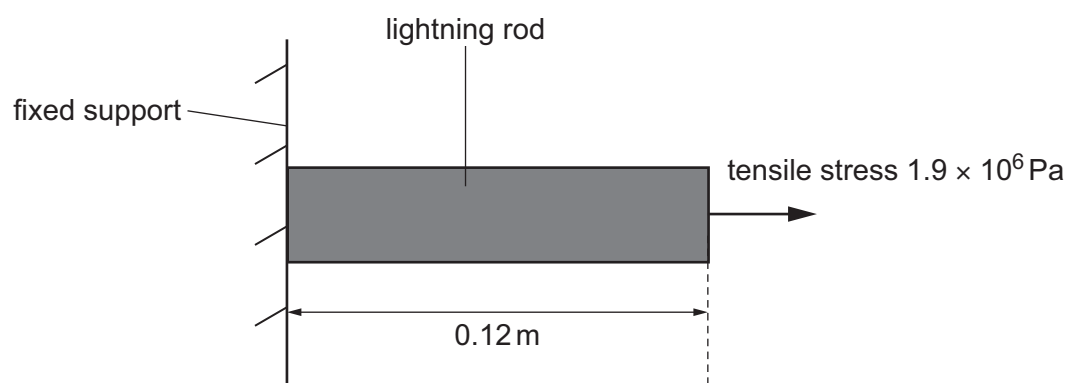


Fig. 3.2 (not to scale)

The section of the rod obeys Hooke's law. The Young modulus of copper is $1.3 \times 10^{11} \text{ Pa}$.

Calculate the extension of the section.

extension = _____ m [3]

17. M/J 24/P22/Q4/a

- 4 A pinball machine uses a spring to launch a small metal ball of mass $4.5 \times 10^{-2} \text{ kg}$ up a ramp. The spring is compressed by $8.0 \times 10^{-2} \text{ m}$ and held in equilibrium, as shown in Fig. 4.1.

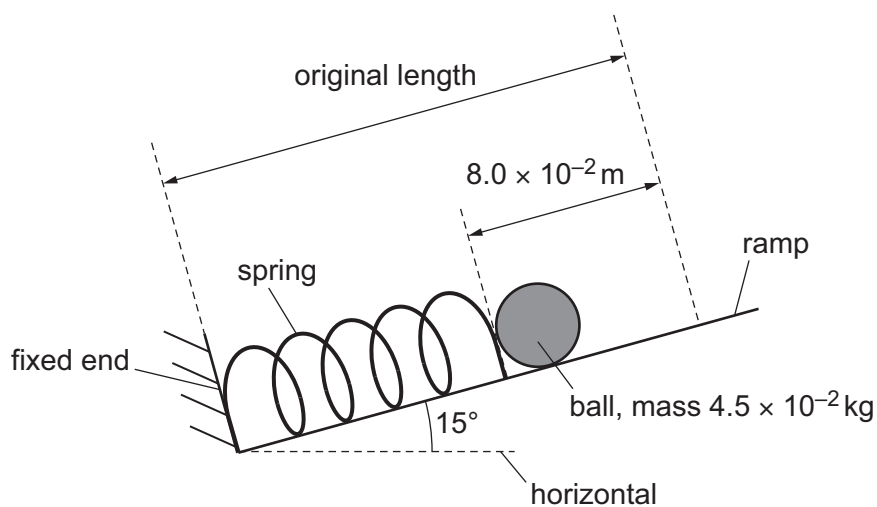


Fig. 4.1 (not to scale)

The ramp is at an angle of 15° to the horizontal.

- (a) The spring obeys Hooke's law and has a spring constant of 29 N m^{-1} .

Calculate the elastic potential energy in the compressed spring.

elastic potential energy =

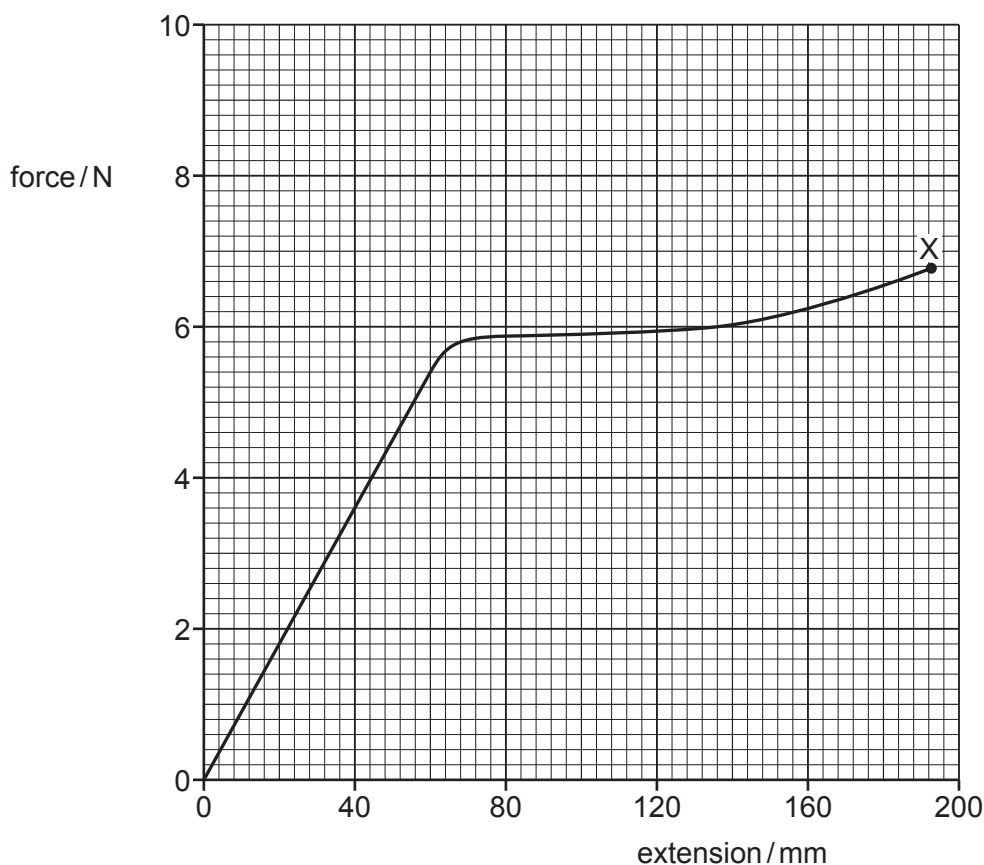
J [2]

18. M/J 24/P23/Q3/a,b,c,d,e

- 3 (a) State Hooke's law.

[1]

- (b) The variation of the applied force with the extension for a sample of a material is shown in Fig. 3.1.

**Fig. 3.1**

The sample behaves elastically up to an extension of 80 mm and breaks at point X.

- (i) On the line in Fig. 3.1, draw a cross (×) to show the limit of proportionality. Label this cross with the letter P. [1]
- (ii) On the line in Fig. 3.1, draw a cross (×) to show the elastic limit. Label this cross with the letter E. [1]
- (c) The sample in (b) has a cross-sectional area of 0.40 mm^2 and an initial length of 3.2 m.
For deformations within the limit of proportionality of the sample, determine:
- (i) the spring constant of the sample
 spring constant = Nm^{-1} [2]
- (ii) the Young modulus of the material from which the sample is made.
 Young modulus = Pa [3]
- (d) Determine an estimate of the work done on the sample as it is extended from zero extension to its breaking point. Explain your reasoning.
 work done = J [2]
- (e) A second sample of the same material has a larger cross-sectional area than the original sample but the same initial length. The two samples are each deformed with the limit of proportionality.

State and explain qualitatively how the spring constant of the second sample compares with that of the original sample.

[2]

19. MAR 24/P22/Q3/a,b

- 3** A thin metal wire X, of diameter $1.2 \times 10^{-3} \text{ m}$, is used to suspend a model planet, as shown in Fig. 3.1.

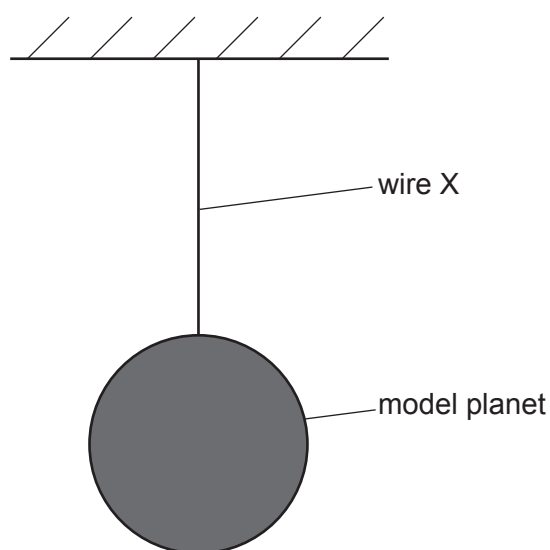


Fig. 3.1 (not to scale)

The variation with strain of the stress for wire X is shown in Fig. 3.2.

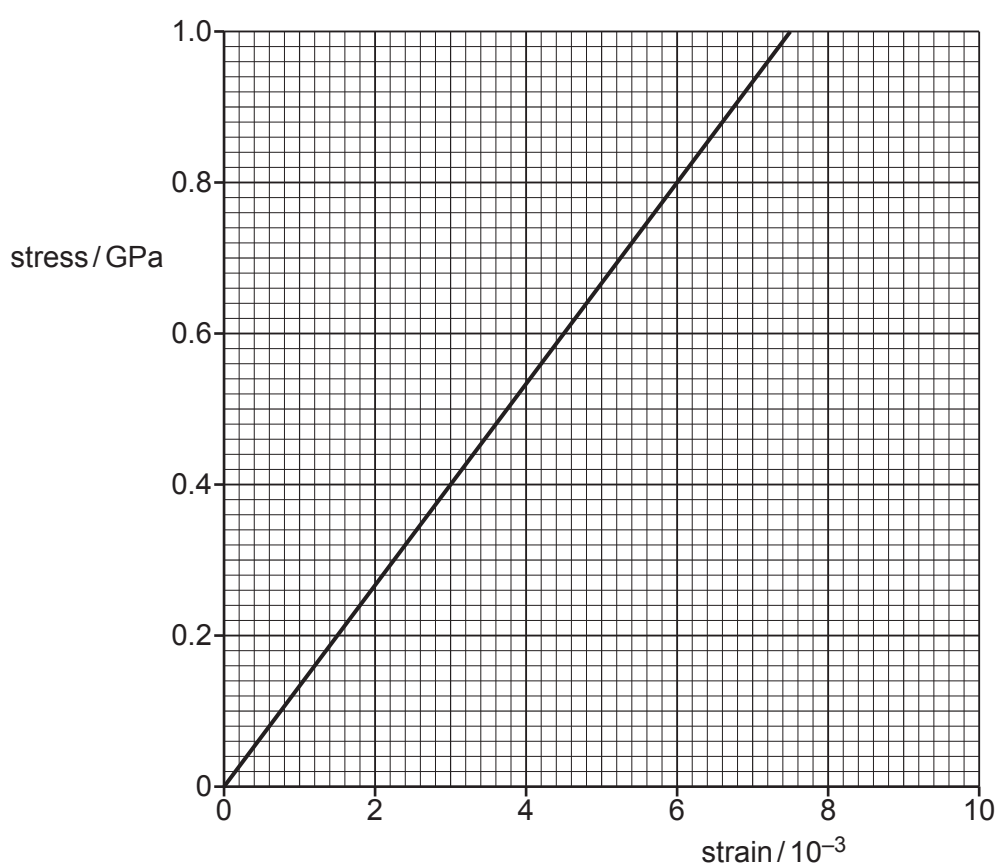


Fig. 3.2

- (a) The strain in X is 5.4×10^{-3} .
- (i) Use Fig. 3.2 to calculate the force exerted on the wire by the model planet.

force = N [3]

- (ii) The elastic potential energy of X is 0.31 J.

Calculate the original length of the wire before the model planet was attached.

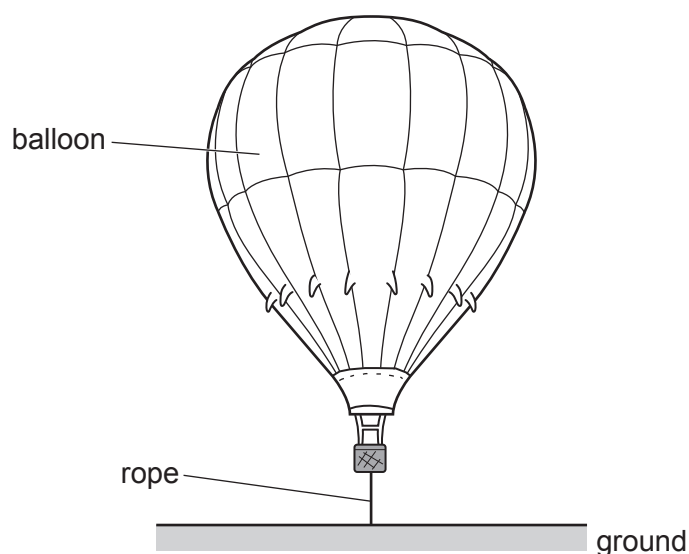
original length = m [3]

- (b) Wire X is replaced by a new wire, Y, with the same original length and diameter but double the Young modulus of X. Wire Y also obeys Hooke's law.

On Fig. 3.2, draw a line representing the variation with strain of the stress for Y. [2]

20. O/N 23/P21/Q2/c

- 2** A hot-air balloon floats just above the ground. The balloon is stationary and is held in place by a vertical rope, as shown in Fig. 2.1.

**Fig. 2.1**

The balloon has a weight W of $3.39 \times 10^4 \text{ N}$. The tension T in the rope is $4.00 \times 10^2 \text{ N}$.
Upthrust U acts on the balloon.
The density of the surrounding air is 1.23 kg m^{-3} .

- (c)** Before the balloon is released, the rope holding the balloon has a strain of 2.4×10^{-5} .
The rope has an unstretched length of 2.5 m . The rope obeys Hooke's law.

- (i)** Show that the extension of the rope is $6.0 \times 10^{-5} \text{ m}$.

[1]

- (ii)** Calculate the elastic potential energy E_p of the rope.

$$E_p = \quad \quad \quad \text{J} \quad [2]$$

- (iii)** The rope holding the balloon is replaced with a new one of the same original length and cross-sectional area. The tension is unchanged and the new rope also obeys Hooke's law.

The new rope is made from a material of a lower Young modulus.

State and explain the effect of the lower Young modulus on the elastic potential energy of the rope.

[2]

21. O/N 23/P22/Q3/a,d

- 3** A vertical rod is fixed to the horizontal surface of a table, as shown in Fig. 3.1.

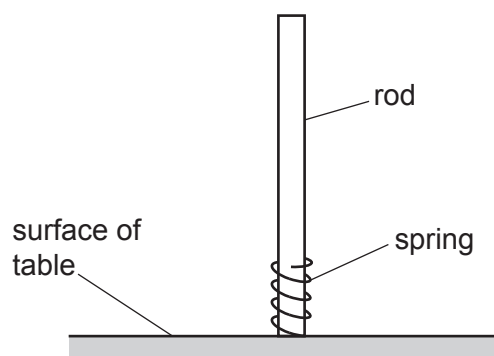


Fig. 3.1 (not to scale)

A spring of mass 7.5 g is able to slide along the full length of the rod.

The spring is first pushed against the surface of the table so that it has an initial compression of 2.1 cm. The spring is then suddenly released so that it leaves the surface of the table with a kinetic energy of 0.048 J and then moves up the rod.

Assume that the spring obeys Hooke's law and that the initial elastic potential energy of the compressed spring is equal to the kinetic energy of the spring as it leaves the surface of the table. Air resistance is negligible.

- (a)** By using the initial elastic potential energy of the compressed spring, calculate its spring constant.

spring constant = Nm^{-1} [2]

- (d)** The rod is replaced by another rod that exerts negligible frictional force on the moving spring. The initial compression x of the spring is now varied in order to vary the maximum increase in height Δh of the spring after leaving the surface of the table. Assume that the spring obeys Hooke's law for all compressions.

On Fig. 3.2, sketch a graph to show the variation with x of Δh . Numerical values are not required.

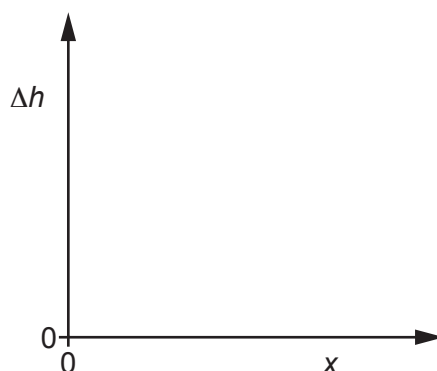
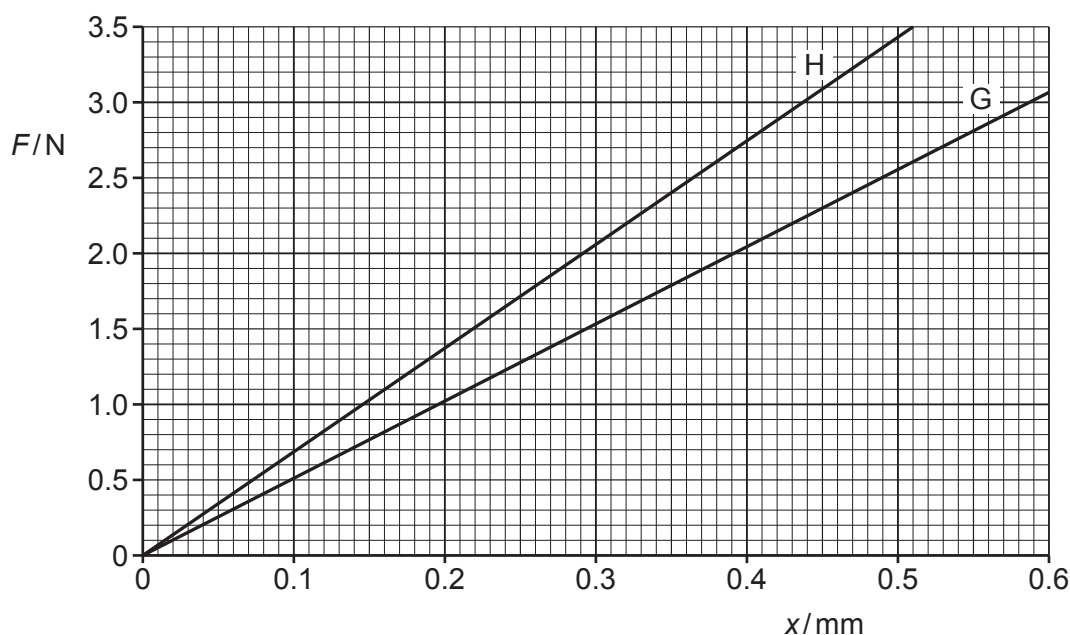


Fig. 3.2

[2]

22. O/N 23/P23/Q4/a,b

- 4 Fig. 4.1 shows the variation with extension x of the tensile force F for two wires, G and H, made from the same material.

**Fig. 4.1**

The elastic limit has not been exceeded for G or H.

- (a) For the lines in Fig. 4.1:

(i) state what is represented by the gradient

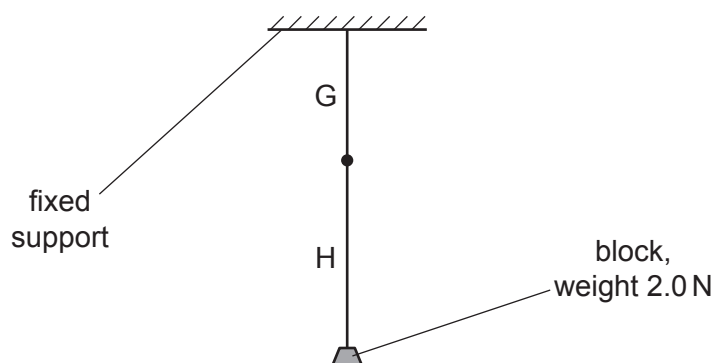
[1]

(ii) explain why the area under the line represents the elastic potential energy of the wire.

[2]

- (b) Wires G and H are joined together end-to-end to form a composite wire of negligible weight. The composite wire hangs vertically from a fixed support.

A block of weight of 2.0 N is attached to the end of the wire, as shown in Fig. 4.2.

**Fig. 4.2**

- (i) Use Fig. 4.1 to determine:

- the extension x_G of wire G

$$x_G = \quad \text{mm}$$

- the extension x_H of wire H.

$$x_H = \quad \text{mm}$$

[1]

- (ii) Calculate the total elastic potential energy E_p of the composite wire due to the weight of the block.

$$E_p = \quad \text{J} \quad [2]$$

- (iii) The original length of wire G is L and the original length of wire H is $1.5L$.

Calculate the ratio

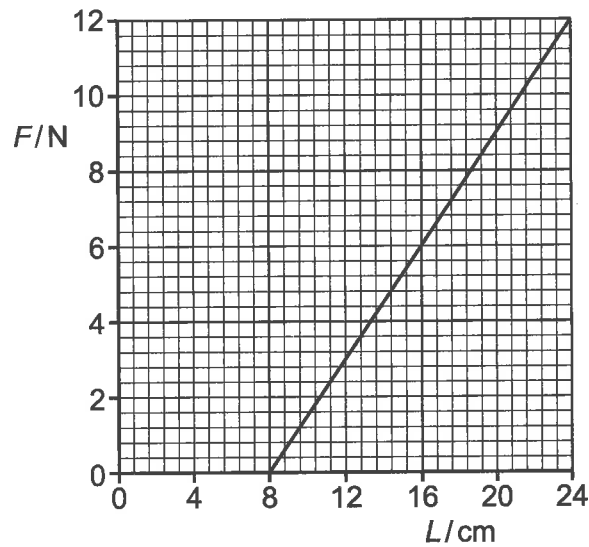
$$\frac{\text{cross-sectional area of wire G}}{\text{cross-sectional area of wire H}}$$

$$\text{ratio} = \quad [3]$$

23. M/J 23/P22/Q4

booklet (scan + OCR)

A spring is suspended from a fixed point at one end. The spring is extended by a vertical force applied to the other end. The variation of the applied force F with the length L of the spring is shown in Fig. 4.1.

**Fig. 4.1**

For the spring:

- (a) state the name of the law that gives the relationship between the force and the extension

..... [1]

- (b) determine the spring constant, in N m^{-1}

spring constant = N m^{-1} [2]

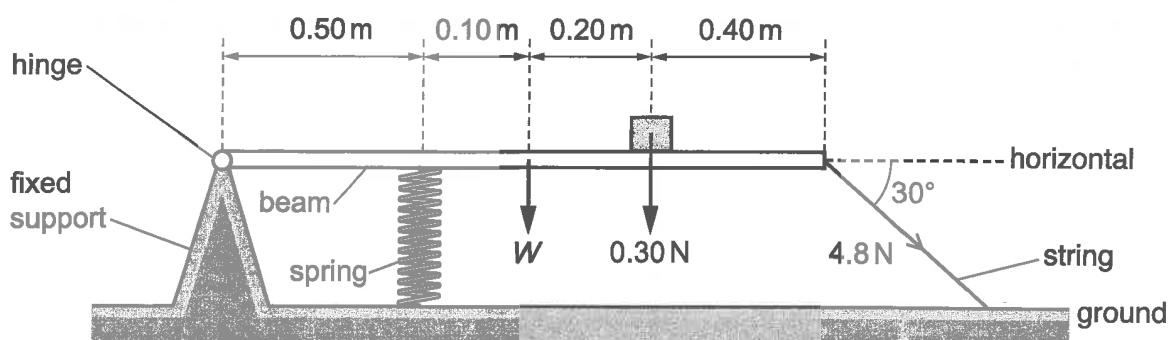
- (c) determine the elastic potential energy when $F = 6.0 \text{ N}$.

elastic potential energy = J [2]

24. M/J 23/P21/Q2/a,b

booklet (scan + OCR)

A rigid uniform beam of weight W is connected to a fixed support by a hinge, as shown in Fig. 2.1.

**Fig. 2.1** (not to scale)

A compressed spring exerts a total force of 8.2 N vertically upwards on the horizontal beam. A block of weight 0.30 N rests on the beam. The right-hand end of the beam is connected to the ground by a string at an angle of 30° to the horizontal. The tension in the string is 4.8 N. The distances along the beam are shown in Fig. 2.1.

The beam is in equilibrium. Assume that the hinge is frictionless.

(a) (i) Show that the vertical component of the tension in the string is 2.4 N.

[1]

(ii) By taking moments about the hinge, determine the weight W of the beam.

$W = \dots\dots\dots$ N [3]

(iii) Calculate the horizontal component of the force exerted on the beam by the hinge.

force = $\dots\dots\dots$ N [1]

(b) The spring obeys Hooke's law and has an elastic potential energy of 0.32 J.

Calculate the compression of the spring.

compression = $\dots\dots\dots$ m [2]

25. M/J 23/P23/Q6

booklet (scan + OCR)

(a) Define the Young modulus.

$\dots\dots\dots$ [1]

- (b) A uniform wire is suspended from a fixed support. Masses are added to the other end of the wire, as shown in Fig. 6.1.

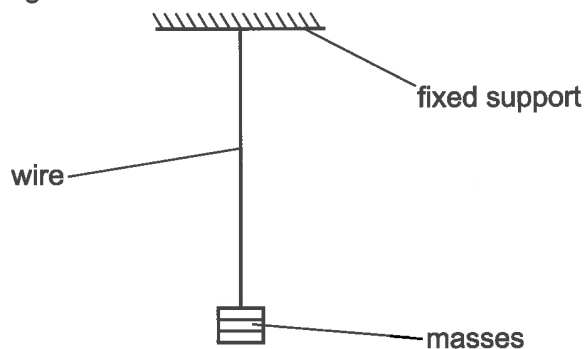


Fig. 6.1 (not to scale)

The variation of the length l of the wire with the force F applied to the wire by the masses is shown in Fig. 6.2.

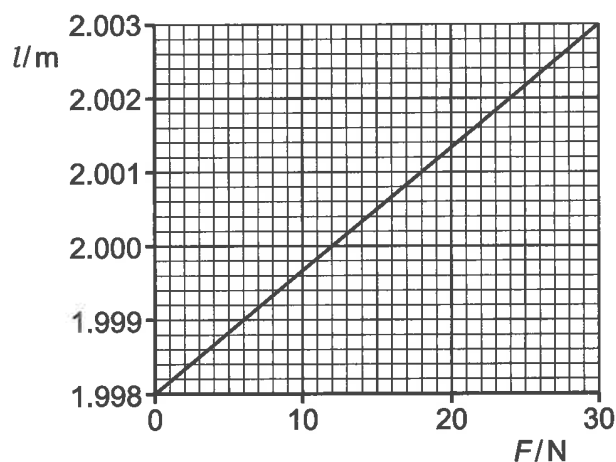


Fig. 6.2

The cross-sectional area of the wire is 0.95 mm^2 .

- (i) Determine the unstretched length of the wire.

unstretched length = m [1]

- (ii) For an applied force F of 30 N, determine:

- the stress in the wire

stress = Pa

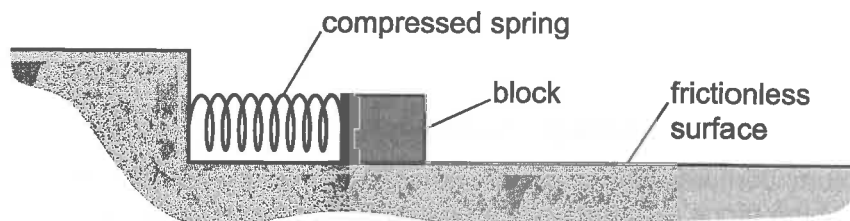
- the strain of the wire.

strain = [3]

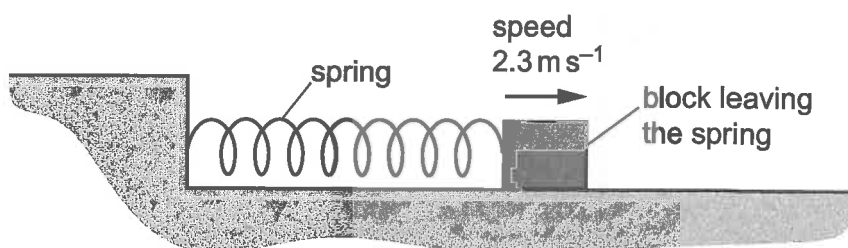
26. O/N 22/P22/Q4

booklet (scan + OCR)

A horizontal spring is fixed at one end. A block is pushed against the other end of the spring so that the spring is compressed, as shown in Fig. 4.1.

**Fig. 4.1**

The block is released and accelerates along a horizontal frictionless surface as the spring returns to its original length. The block leaves the end of the spring with a speed of 2.3 m s^{-1} , as shown in Fig. 4.2.

**Fig. 4.2**

The block has a mass of 250 g and the spring has a spring constant of 420 N m^{-1} .

Assume that the spring always obeys Hooke's law and that all the elastic potential energy of the spring is transferred to the kinetic energy of the block.

(a) Calculate the kinetic energy of the block as it leaves the spring.

kinetic energy = J [2]

(b) Calculate the compression of the spring immediately before the block is released.

compression = m [2]

(c) After leaving the spring, the block moves along the surface until it hits a barrier at a speed of 2.3 m s^{-1} . The block then rebounds at a speed of 1.5 m s^{-1} and moves back along its original path. The block is in contact with the barrier for a time of 0.086 s.

Calculate:

(i) the change in momentum of the block during the collision

change in momentum = N s [2]

- (ii) the average resultant force exerted on the block during the collision.

average resultant force = N [1]

- (d) The maximum compression x of the spring is now varied in order to vary the kinetic energy E_K of the block as it leaves the spring. Assume that all the elastic potential energy in the spring is always transferred to the kinetic energy of the block.

On Fig. 4.3, sketch a graph to show the variation with x of E_K .

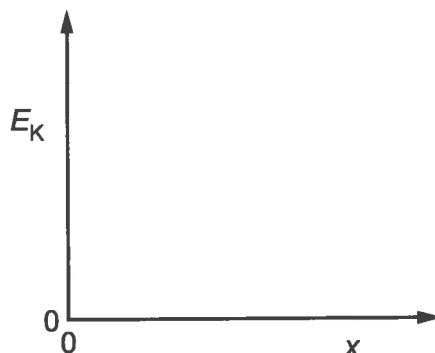


Fig. 4.3

[1]

27. O/N 22/P21/Q2

booklet (scan + OCR)

A steel ball is projected horizontally from the top of a table, as shown in Fig. 2.1.

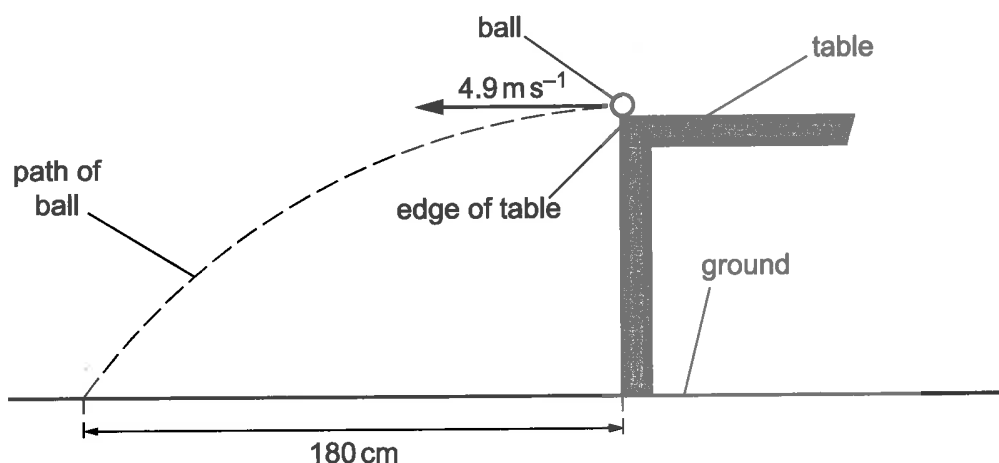


Fig. 2.1 (not to scale)

The ball is projected horizontally at a speed of 4.9 ms^{-1} . The ball lands on the ground a horizontal distance of 180 cm from the edge of the table.

Assume that air resistance is negligible.

- (a) (i) Calculate the time taken for the ball to reach the ground.

time = s [1]

- (ii) Calculate the vertical component of the velocity of the ball as it hits the ground.

velocity = ms^{-1} [2]

- (iii) Determine the magnitude and the angle to the horizontal of the velocity of the ball as it hits the ground.

magnitude of velocity = ms^{-1}

angle to the horizontal = $^{\circ}$ [3]

- (b) The ball is projected by means of a compressed spring which is attached to a fixed block as shown in Fig. 2.2.

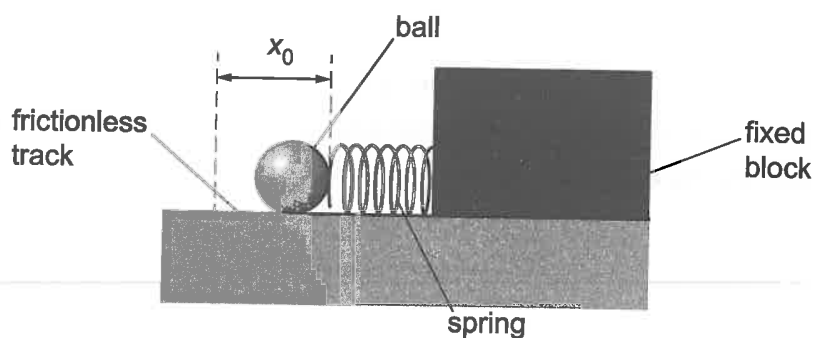


Fig. 2.2

The ball is placed on a frictionless track in front of the spring. The ball is then pulled back so that the spring has compression x_0 .

When the spring is released, the ball is projected horizontally as shown in Fig. 2.3.

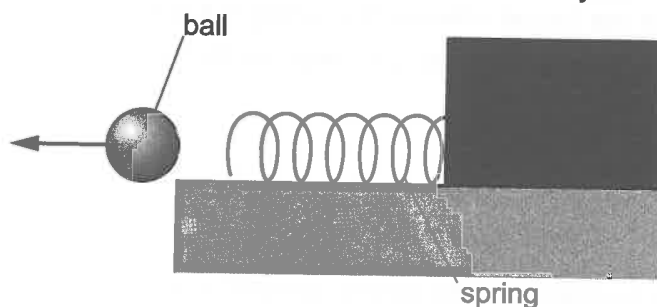


Fig. 2.3

The variation with compression x of the applied force F for the spring is shown in Fig. 2.4.

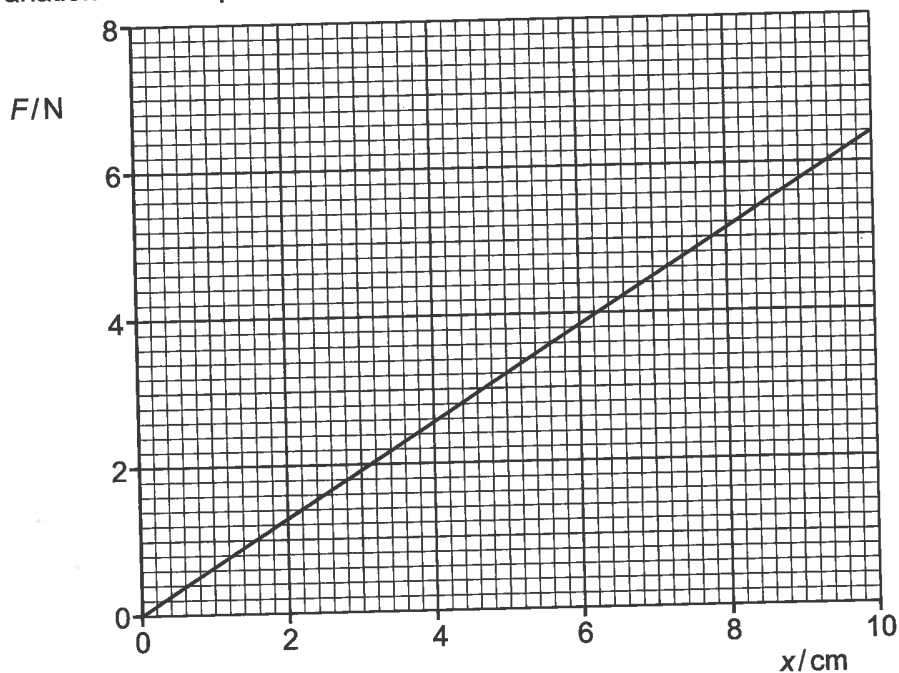


Fig. 2.4

The ball is a uniform sphere of steel of diameter 0.016 m and mass 0.017 kg.

- (i) Calculate the density of the steel.

density = kg m^{-3} [3]

- (ii) All of the elastic potential energy in the spring is converted into kinetic energy of the ball. The speed of the ball as it leaves the spring is 4.9 m s^{-1} . Show that the maximum elastic potential energy of the spring is 0.20 J.

[2]

- (iii) Use Fig. 2.4 to determine the spring constant k of the spring.

k = N m^{-1} [2]

- (iv) Use your answer in (b)(iii) and the value of energy given in (b)(ii) to determine the compression x_0 of the spring.

$$x_0 = \dots\dots\dots \text{ m [2]}$$

- (c) The steel ball is replaced by a polystyrene ball of the same diameter but of much lower mass. The spring is given compression x_0 and is then released.

Air resistance on this ball is **not** negligible after it leaves the spring.

Explain:

- (i) why this ball leaves the spring with a greater speed than that of the steel ball

..... [1]

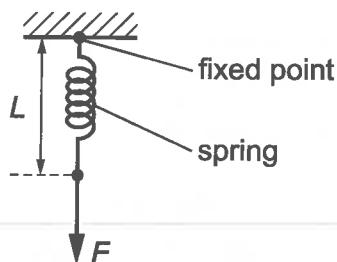
- (ii) why this ball takes a longer time to reach the ground than the steel ball.

..... [1]

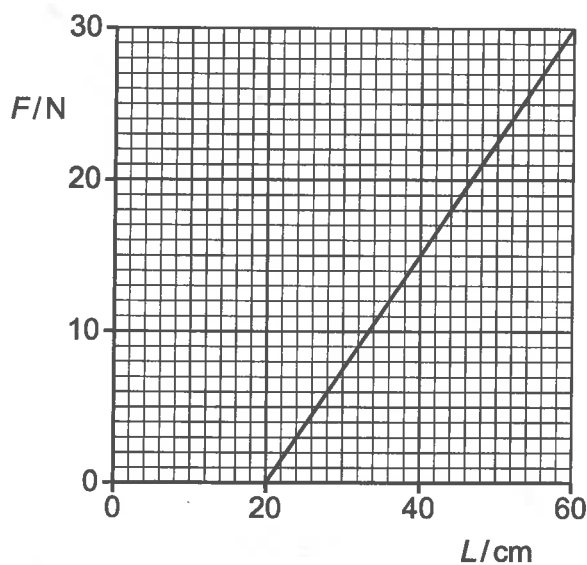
28. M/J 22/P21/Q4

booklet (scan + OCR)

A spring is suspended from a fixed point at one end and a vertical force is applied to the other end, as shown in Fig. 4.1.

**Fig. 4.1**

The variation of the applied force F with the length L of the spring is shown in Fig. 4.2.

**Fig. 4.2**

- (a) Determine the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \text{ [2]}$$

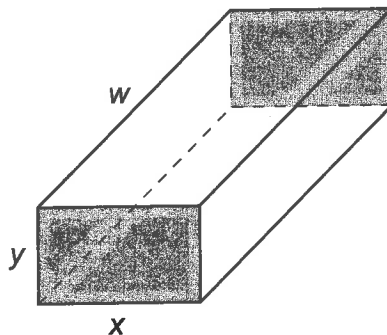
- (b) Determine the elastic potential energy in the spring when the applied force F is 15 N.

$$\text{elastic potential energy} = \dots\dots\dots \text{J [3]}$$

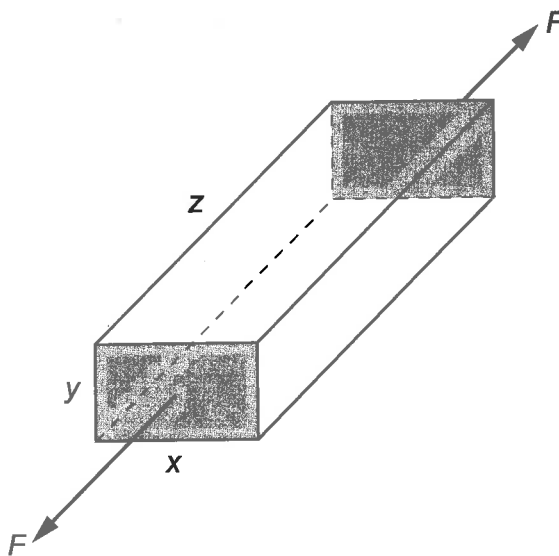
29. O/N 21/P21/Q3

booklet (scan + OCR)

(a) A uniform metal bar, initially unstretched, has sides of length w , x and y , as shown in Fig. 3.1.

**Fig. 3.1**

The bar is now stretched by a tensile force F applied to the shaded ends. The changes in the lengths x and y are negligible. The bar now has sides of length x , y and z , as shown in Fig. 3.2.

**Fig. 3.2**

Determine expressions, in terms of some or all of F , w , x , y and z , for:

(i) the stress σ applied to the bar by the tensile force

$$\sigma = \dots\dots\dots [1]$$

(ii) the strain ε in the bar due to the tensile force

$$\varepsilon = \dots\dots\dots [1]$$

(iii) the Young modulus E of the metal from which the bar is made.

$$E = \dots\dots\dots [2]$$

(b) A copper wire is stretched by a tensile force that gradually increases from 0 to 280 N. The variation with extension of the tensile force is shown in Fig. 3.3.

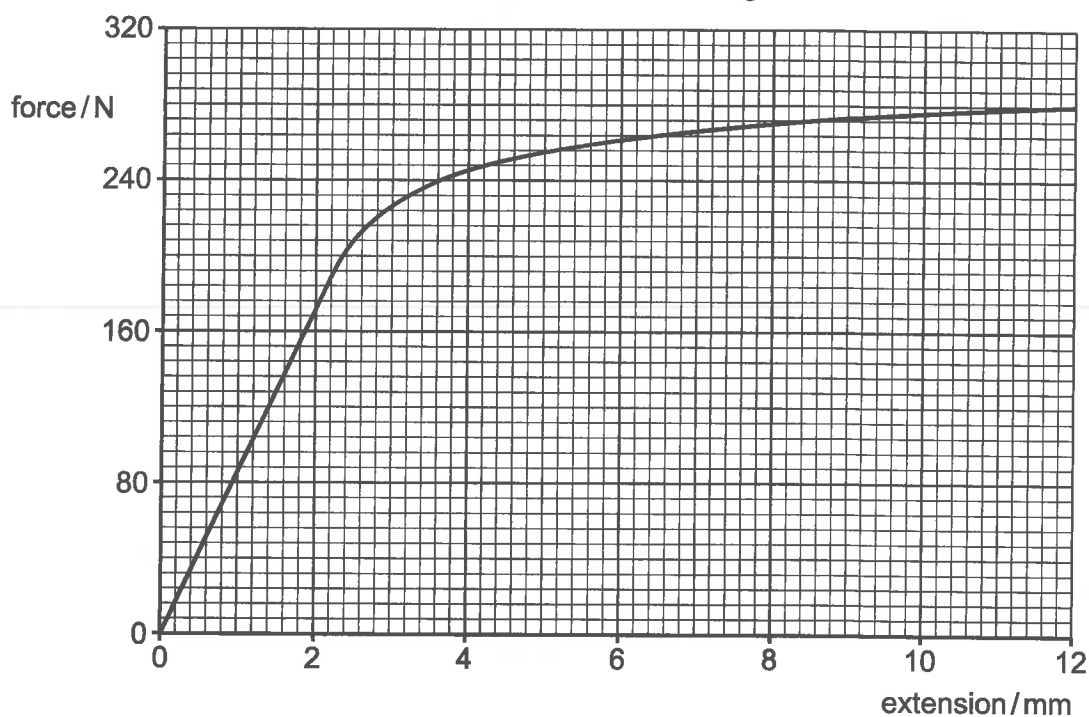


Fig. 3.3

(i) State the maximum extension of the wire for which it obeys Hooke's law.

$$\text{extension} = \dots\dots\dots \text{ mm } [1]$$

(ii) Use Fig. 3.3 to determine the strain energy in the wire when the tensile force is 120 N.

$$\text{strain energy} = \dots\dots\dots \text{ J } [3]$$

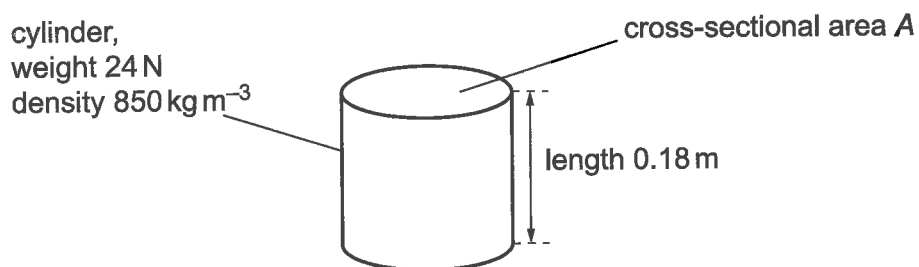
(iii) Explain why the work done in stretching the wire to an extension of 12 mm is not equal to the energy recovered when the tensile force is removed.

.....
 [2]

30. O/N 21/P23/Q1

booklet (scan + OCR)

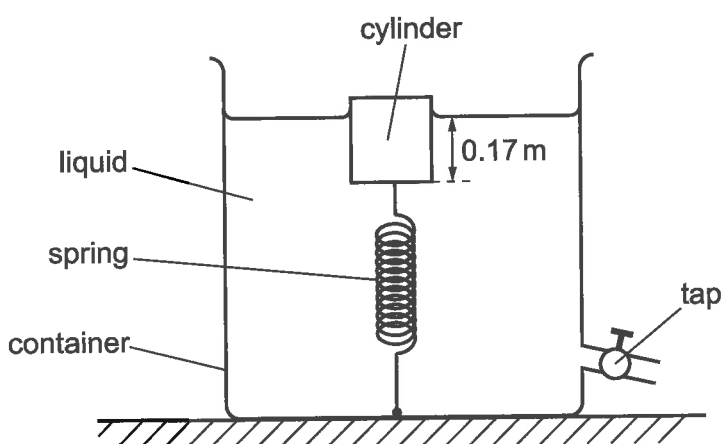
- (a) A solid cylinder of weight 24 N is made of material of density 850 kg m^{-3} . The cylinder has a length of 0.18 m, as shown in Fig. 1.1.

**Fig. 1.1**

Show that the cross-sectional area A of the cylinder is 0.016 m^2 .

[3]

- (b) The cylinder in (a) is attached by a spring to the bottom of a rigid container of liquid, as shown in Fig. 1.2.

**Fig. 1.2 (not to scale)**

The cylinder is in equilibrium with its bottom face at a depth of 0.17 m below the surface of the liquid. The tension in the spring is 8.0 N.

(i) Show that the upthrust acting on the cylinder due to the liquid is 32 N.

[1]

(ii) Calculate the density of the liquid.

density = kg m^{-3} [3]

(c) Fig. 1.3 shows the variation of the tension F with the length of the spring in (b).

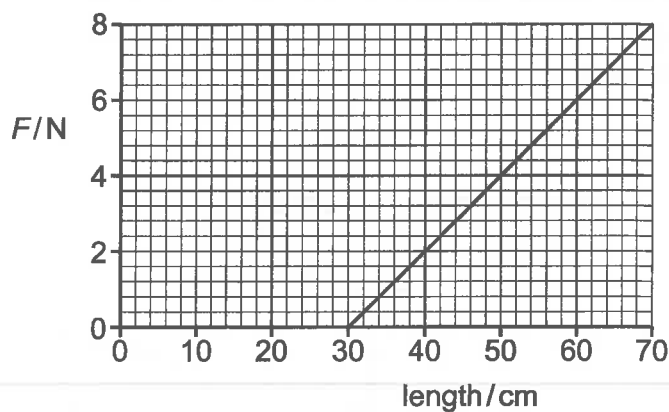


Fig. 1.3

(i) The tap at the bottom of the container is opened so that a fixed amount of liquid flows out of the container. The cylinder moves downwards so that the tension in the spring changes from 8.0 N to 4.0 N.

Determine the change in the elastic potential energy of the spring.

change in elastic potential energy = J [3]

(ii) More liquid is let out of the container until the upthrust on the cylinder becomes 24 N. For the upthrust of 24 N, determine the length of the spring.

length = cm [1]

31. M/J 21/P22/Q1

booklet (scan + OCR)

(a) Complete Table 1.1 by stating whether each of the quantities is a vector or a scalar.

Table 1.1

quantity	vector or scalar
acceleration	
electrical resistance	
momentum	

[2]

(b) State the conditions for an object to be in equilibrium.

.....

..... [2]

(c) A floating solid cylinder is attached by a wire to the sea bed, as shown in Fig. 1.1.

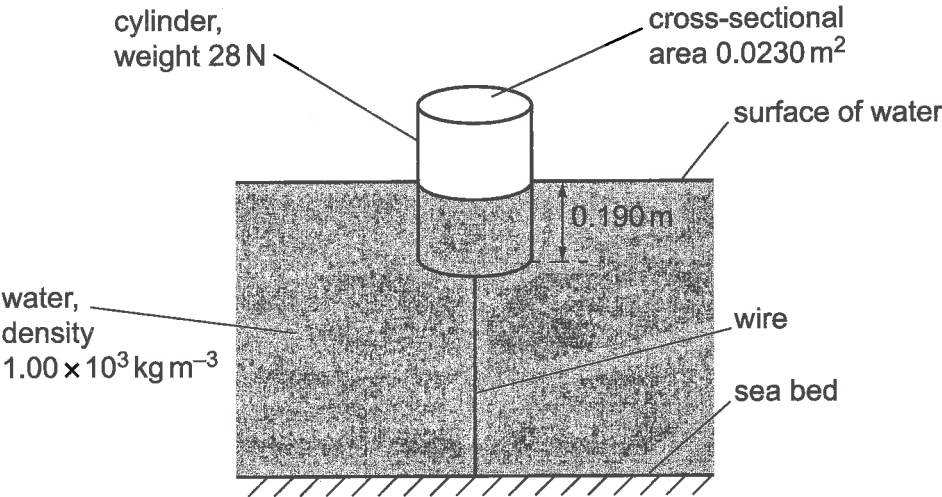


Fig. 1.1 (not to scale)

The density of the water is $1.00 \times 10^3 \text{ kg m}^{-3}$. The base of the cylinder is at a depth of 0.190 m below the surface of the water. The cylinder has a weight of 28 N and a cross-sectional area of 0.0230 m^2 .

The wire and the central axis of the cylinder are both vertical. The cylinder is in equilibrium.

(i) Calculate, to three significant figures, the upthrust acting on the cylinder due to the water.

upthrust = N [2]

(ii) Show that the tension T in the wire is 15 N.

[1]

- (iii) The wire has a cross-sectional area of 3.2 mm^2 .
Calculate the stress in the wire.

stress = Pa [2]

- (iv) The surface of the water gradually rises until it is level with the top face of the cylinder.
State and explain, qualitatively, the variation of the strain energy stored in the wire as the water surface rises.

.....
..... [2]

32. M/J 21/P23/Q3

booklet (scan + OCR)

- (a) Define the *moment* of a force about a point.

.....
..... [2]

- (b) Fig. 3.1 shows a type of balance that is used for measuring mass.

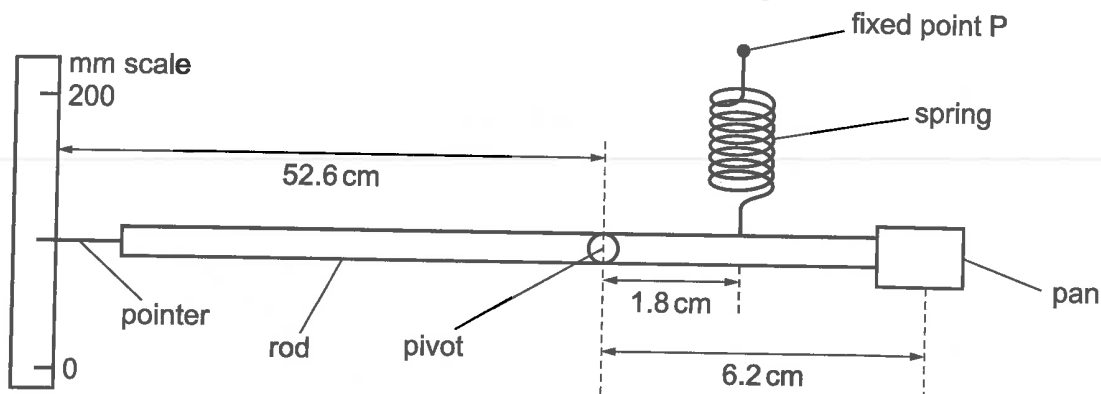


Fig. 3.1 (not to scale)

A rigid rod is pivoted about a point 6.2 cm from the centre of a pan which is attached to one end. The object being measured is placed on the centre of this pan.

A spring, attached to the rod 1.8 cm from the pivot, is attached at its other end to a fixed point P. The spring obeys Hooke's law over the full range of operation of the balance.

A pointer, on the other side of the pivot, is set against a millimetre scale which is a distance 52.6 cm from the pivot.

When the system is in equilibrium with no mass on the pan, the rod is horizontal and the pointer indicates a reading on the scale of 86 mm .

An object of mass 0.472 kg is now placed on the pan. As a result, the pointer moves to indicate a reading of 123 mm on the scale when the system is again in equilibrium.

- (i) Show that the increase in the length of the spring is approximately 1.3 mm .

[2]

- (ii) Calculate the magnitude of the moment about the pivot of the weight of the object.

moment = Nm [2]

- (iii) Use your answer in (b)(ii) to determine the increase in the tension in the spring due to the 0.472 kg mass.

increase in tension = N [2]

- (iv) Use the information in (b)(i) and your answer in (b)(iii) to determine the spring constant k of the spring. Give a unit with your answer.

k = unit [2]

33. MAR 21/P22/Q3

booklet (scan + OCR)

A spring is extended by a force. The variation with extension x of the force F is shown in Fig. 3.1.

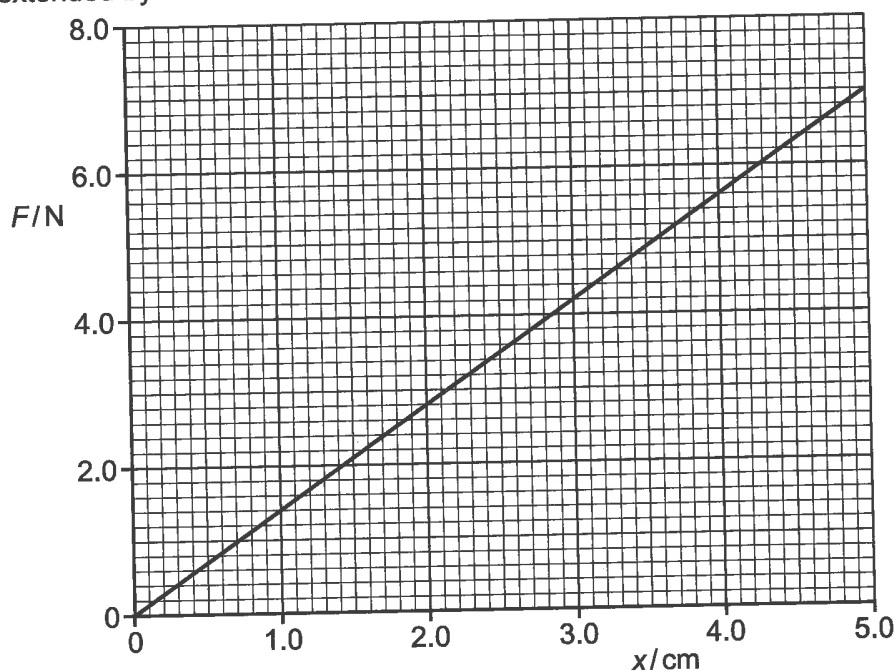


Fig. 3.1

- (a) State the name of the law that relates the force and extension of the spring shown in Fig. 3.1.

..... [1]

(b) Determine:

(i) the spring constant, in N m^{-1} , of the spring

spring constant = N m^{-1} [2]

(ii) the strain energy (elastic potential energy) in the spring when the extension is 4.0 cm.

strain energy = J [2]

(c) One end of the spring is attached to a fixed point. A cylinder that is submerged in a liquid is now suspended from the other end of the spring, as shown in Fig. 3.2.

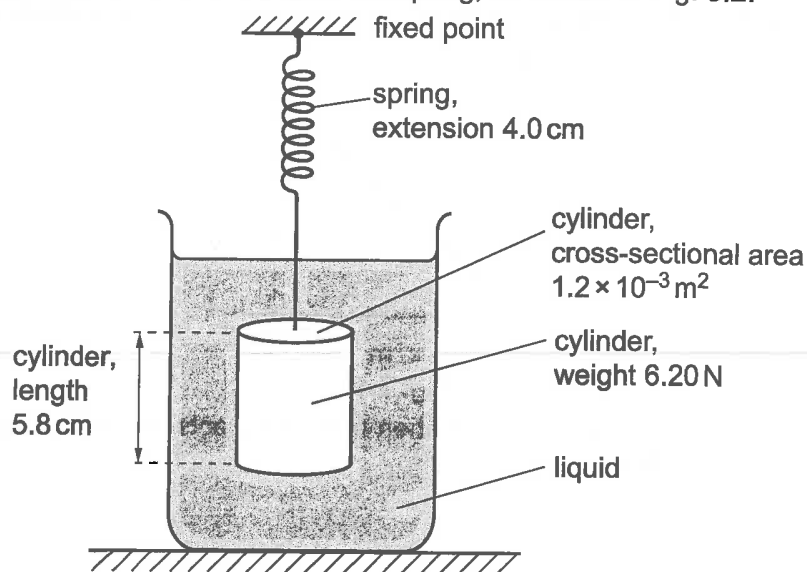


Fig. 3.2

The cylinder has length 5.8 cm, cross-sectional area $1.2 \times 10^{-3} \text{ m}^2$ and weight 6.20 N. The cylinder is in equilibrium when the extension of the spring is 4.0 cm.

(i) Show that the upthrust acting on the cylinder is 0.60 N.

[1]

(ii) Calculate the difference in pressure between the bottom face and the top face of the cylinder.

difference in pressure = Pa [2]

(iii) Calculate the density of the liquid.

density = kg m^{-3} [2]

(d) The liquid in (c) is replaced by another liquid of greater density.

State the effect, if any, of this change on:

(i) the upthrust acting on the cylinder

..... [1]

(ii) the extension of the spring.

..... [1]

34. O/N 20/P22/Q3

booklet (scan + OCR)

(a) A spring is fixed at one end and is compressed by applying a force to the other end. The variation of the force F acting on the spring with its compression x is shown in Fig. 3.1.

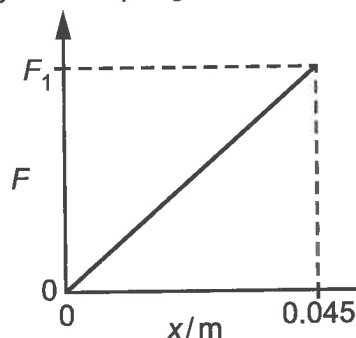


Fig. 3.1

A compression of 0.045 m is produced when a force F_1 acts on the spring. The spring has a spring constant of 800 N m^{-1} .

(i) Determine F_1 .

$F_1 = \dots\dots\dots \text{ N}$ [2]

(ii) Use Fig. 3.1 to show that, for a compression of 0.045 m, the elastic potential energy of the spring is 0.81 J.

[2]

(b) A child's toy uses the spring in (a) to launch a ball of mass 0.020 kg vertically into the air. The ball is initially held against one end of the spring which has a compression of 0.045 m. The spring is then released to launch the ball. The kinetic energy of the ball as it leaves the toy is 0.72 J.

(i) The toy converts the elastic potential energy of the spring into the kinetic energy of the

ball. Use the information in (a)(ii) to calculate the percentage efficiency of this conversion.

efficiency = % [1]

(ii) Determine the initial momentum of the ball as it leaves the toy.

momentum = N s [3]

(c) The ball in (b) leaves the toy at point A and moves vertically upwards through the air. Point B is the position of the ball when it is at maximum height h above point A, as illustrated in Fig. 3.2.

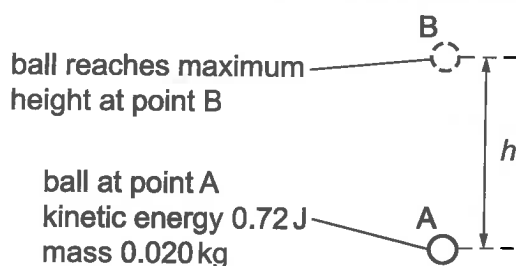


Fig. 3.2 (not to scale)

The gravitational potential energy of the ball increases by 0.60 J as it moves from A to B.

(i) Calculate h .

$h =$ m [2]

(ii) Determine the average force due to air resistance acting on the ball for its movement from A to B.

average force = N [2]

(iii) When there is air resistance, the ball takes time T to move from A to B.

State and explain whether the time taken for the ball to move from A to its maximum height will be more than, less than or equal to time T if there is no air resistance.

..... [1]

35. O/N 20/P21/Q4

booklet (scan + OCR)

(a) Define, for a wire:

(i) stress

..... [1]

(ii) *strain*.

..... [1]

- (b) (i) A school experiment is performed on a metal wire to determine the Young modulus of the metal. A force is applied to one end of the wire which is fixed at the other end. The variation of the force F with extension x of the wire is shown in Fig. 4.1.

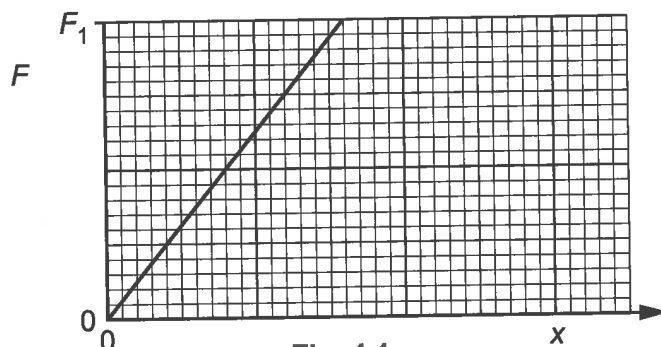


Fig. 4.1

The maximum force applied to the wire is F_1 .

The gradient of the graph line in Fig. 4.1 is G . The wire has initial length L and cross-sectional area A .

Determine an expression, in terms of A , G and L , for the Young modulus E of the metal.

$E =$ [2]

- (ii) A student repeats the experiment in (b)(i) using a new wire that has twice the diameter of the first wire. The initial length of the wire and the metal of the wire are unchanged. On Fig. 4.1, draw the graph line representing the new wire for the force increasing from $F = 0$ to $F = F_1$. [2]

- (iii) Another student repeats the original experiment in (b)(i), increasing the force beyond F_1 to a new maximum force F_2 . The new graph obtained is shown in Fig. 4.2.

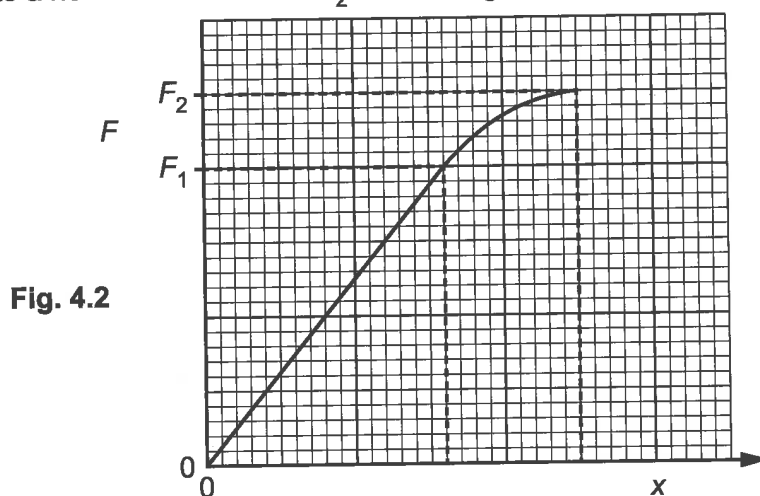


Fig. 4.2

- On Fig. 4.2, shade an area that represents the work done to extend the wire when the force is increased from F_1 to F_2 . [1]
- Explain how the student can check that the elastic limit of the wire was not exceeded when force F_2 was applied.

..... [1]

- (iv) Each student in the class performs the experiment in (b)(i). The teacher describes the values of the Young modulus calculated by the students as having high accuracy and low precision.

Explain what is meant by *low precision*.

..... [1]

36. O/N 20/P23/Q4

booklet (scan + OCR)

- (a) State Hooke's law.

..... [1]

- (b) A spring is fixed at one end. A compressive force F is applied to the other end. The variation of the force F with the compression x of the spring is shown in Fig. 4.1.

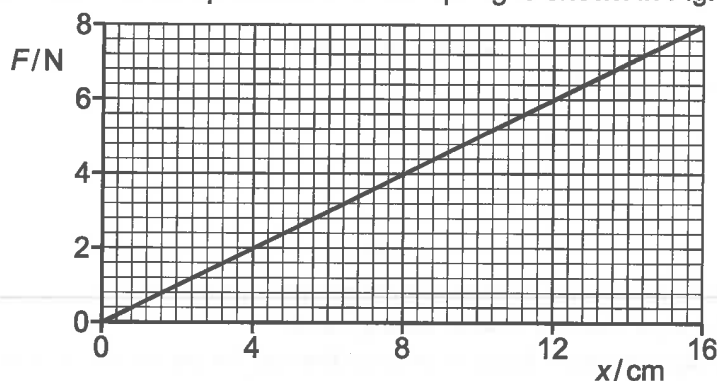


Fig. 4.1

Show that the elastic potential energy of the spring is 0.64 J when its compression is 16.0 cm.

[2]

- (c) The spring in (b) is used to project a toy car along a track from point X to point Y, as illustrated in Fig. 4.2.

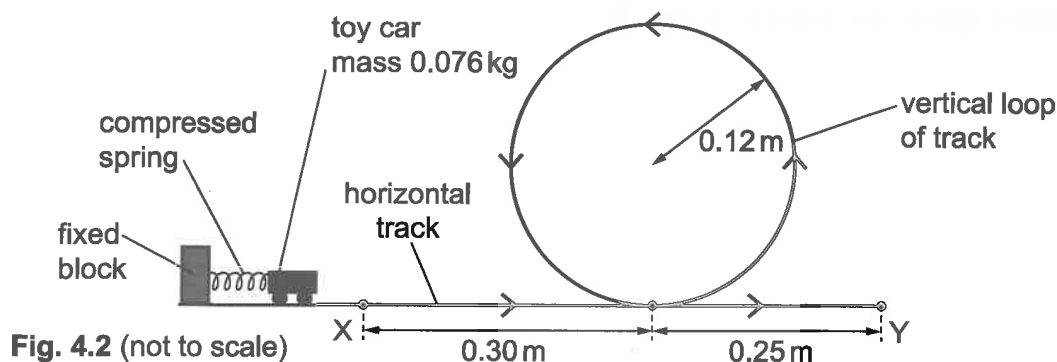


Fig. 4.2 (not to scale)

The spring is initially given a compression of 16.0 cm. The car of mass 0.076 kg is held against one end of the compressed spring. When the spring is released it projects the car forward. The car leaves the spring at point X with kinetic energy that is equal to the initial elastic potential energy of the compressed spring.

The car follows the track around a vertical loop of radius 0.12 m and then passes point Y. Assume that friction and air resistance are negligible.

Calculate:

(i) the speed of the car at X

speed = ms^{-1} [2]

(ii) the kinetic energy of the car when it is at the top of the loop

kinetic energy = J [3]

(iii) the speed of the car at Y.

speed = ms^{-1} [1]

(d) In practice, a resistive force due to friction and air resistance acts on the car so that its kinetic energy at Y is 0.23 J less than its kinetic energy at X.

Determine the average resistive force acting on the car for its movement from X to Y.

average resistive force = N [3]

37. M/J 20/P22/Q5

booklet (scan + OCR)

One end of a wire is attached to a fixed point. A force F is applied to the wire to cause extension x . The variation with F of x is shown in Fig. 5.1.

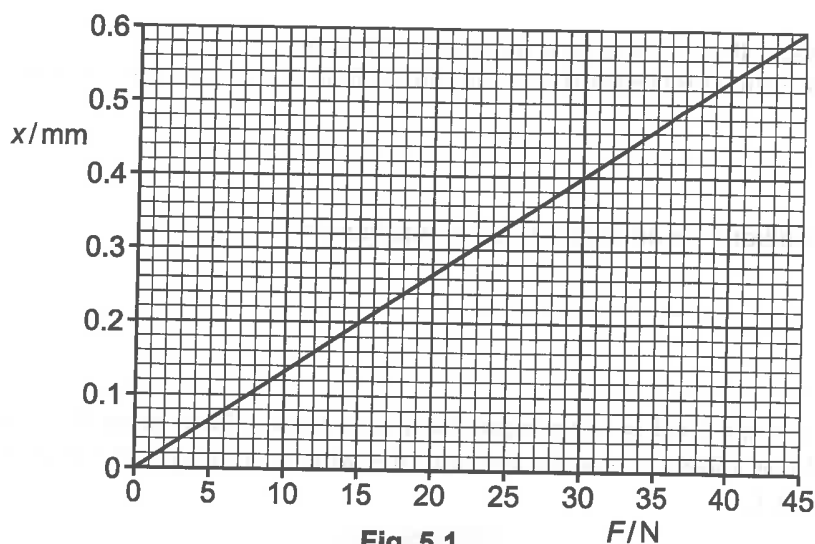


Fig. 5.1

The wire has a cross-sectional area of $4.1 \times 10^{-7} \text{ m}^2$ and is made of metal of Young modulus $1.7 \times 10^{11} \text{ Pa}$. Assume that the cross-sectional area of the wire remains constant as the wire extends.

(a) State the name of the law that describes the relationship between F and x shown in Fig. 5.1.

..... [1]

(b) The wire has an extension of 0.48 mm.

Determine:

(i) the stress

stress = Pa [2]

(ii) the strain.

strain = [2]

(c) The resistivity of the metal of the wire is $3.7 \times 10^{-7} \Omega \text{ m}$.

Determine the change in resistance of the wire when the extension x of the wire changes from $x = 0.48 \text{ mm}$ to $x = 0.60 \text{ mm}$.

change in resistance = Ω [3]

(d) A force of greater than 45 N is now applied to the wire.
Describe how it may be checked that the elastic limit of the wire has not been exceeded.

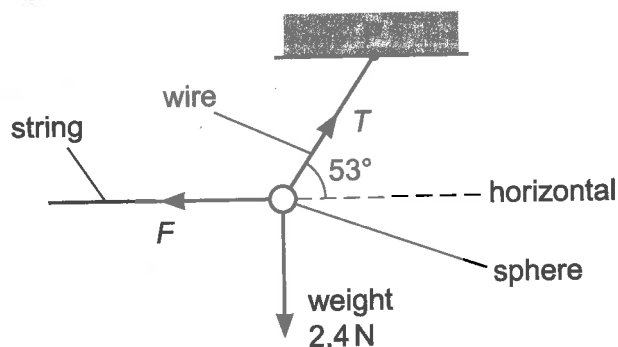
..... [1]

38. M/J 20/P21/Q3

booklet (scan + OCR)

(a) State **two** conditions for an object to be in equilibrium.

1. [2]
2.

(b) A sphere of weight 2.4 N is suspended by a wire from a fixed point P. A horizontal string is used to hold the sphere in equilibrium with the wire at an angle of 53° to the horizontal, as shown in Fig. 3.1.**Fig. 3.1** (not to scale)**(i)** Calculate:

1. the tension
- T
- in the wire

$$T = \dots\dots\dots \text{N}$$

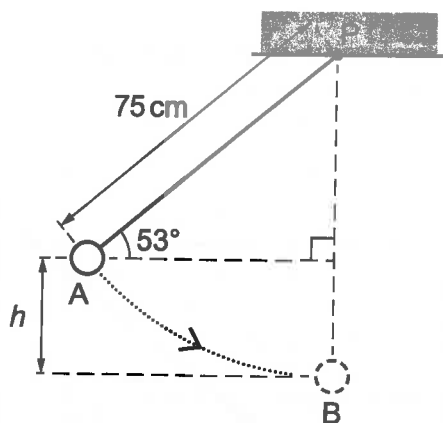
2. the force
- F
- exerted by the string on the sphere.

$$F = \dots\dots\dots \text{N} [2]$$

(ii) The wire has a circular cross-section of diameter 0.50 mm. Determine the stress σ in the wire.

$$\sigma = \dots\dots\dots \text{Pa} [3]$$

(c) The string is disconnected from the sphere in **(b)**. The sphere then swings from its initial rest position A, as illustrated in Fig. 3.2.

**Fig. 3.2** (not to scale)

The sphere reaches maximum speed when it is at the bottom of the swing at position B. The distance between P and the centre of the sphere is 75 cm.

Air resistance is negligible and energy losses at P are negligible.

(i) Show that the vertical distance h between A and B is 15 cm.

[1]

(ii) Calculate the change in gravitational potential energy of the sphere as it moves from A to B.

change in gravitational potential energy = J [2]

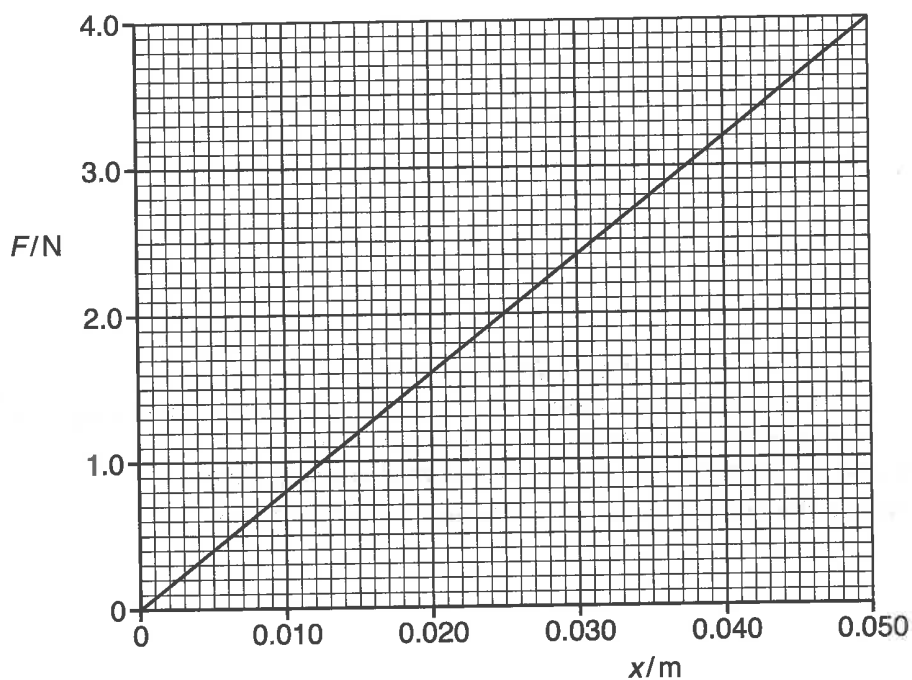
(iii) Use your answer in (c)(ii) to determine the speed of the sphere at B.
Show your working.

speed = ms^{-1} [3]

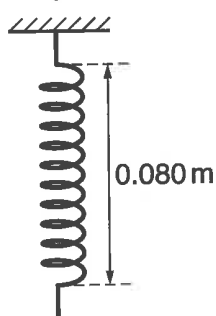
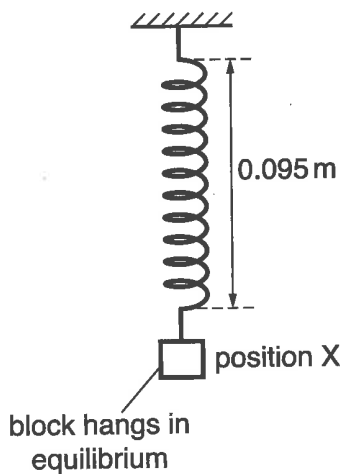
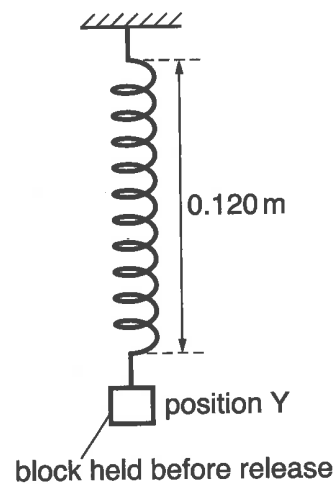
39. O/N 19/P21/Q4

booklet (scan + OCR)

The variation with extension x of the force F applied to a spring is shown in Fig. 4.1.

**Fig. 4.1**

The spring has an unstretched length of 0.080 m and is suspended vertically from a fixed point, as shown in Fig. 4.2.

**Fig. 4.2****Fig. 4.3****Fig. 4.4**

A block is attached to the lower end of the spring. The block hangs in equilibrium at position X when the length of the spring is 0.095 m, as shown in Fig. 4.3.

The block is then pulled vertically downwards and held at position Y so that the length of the spring is 0.120 m, as shown in Fig. 4.4. The block is then released and moves vertically upwards from position Y back towards position X.

(a) Use Fig. 4.1 to determine the spring constant of the spring.

spring constant = Nm^{-1} [2]

- (b) Use Fig. 4.1 to show that the decrease in elastic potential energy of the spring is 0.055 J when the block moves from position Y to position X.

[2]

- (c) The block has a mass of 0.122 kg . Calculate the increase in gravitational potential energy of the block for its movement from position Y to position X.

increase in gravitational potential energy = J [2]

- (d) Use the decrease in elastic potential energy stated in (b) and your answer in (c) to determine, for the block, as it moves through position X:

- (i) its kinetic energy

kinetic energy = J [1]

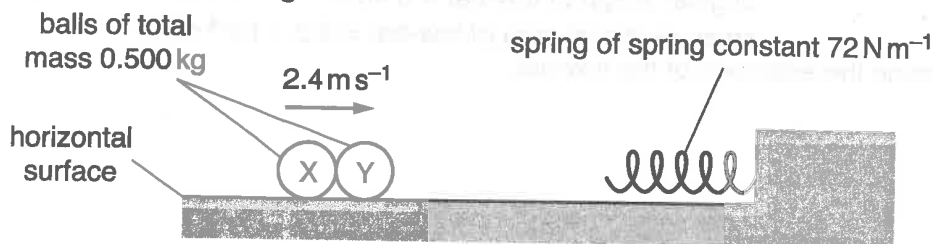
- (ii) its speed.

speed = ms^{-1} [2]

40. O/N 19/P23/Q4/b

booklet (scan + OCR)

- (a)** The two balls continue moving together along the horizontal frictionless surface towards a spring, as illustrated in Fig. 4.3.

**Fig. 4.3**

The balls hit the spring and remain stuck together as they decelerate to rest. All the kinetic energy of the balls is converted into elastic potential energy of the spring. The energy E stored in the spring is given by

$$E = \frac{1}{2} kx^2$$

where k is the spring constant of the spring and x is its compression. The spring obeys Hooke's law and has a spring constant of 72 N m^{-1} .

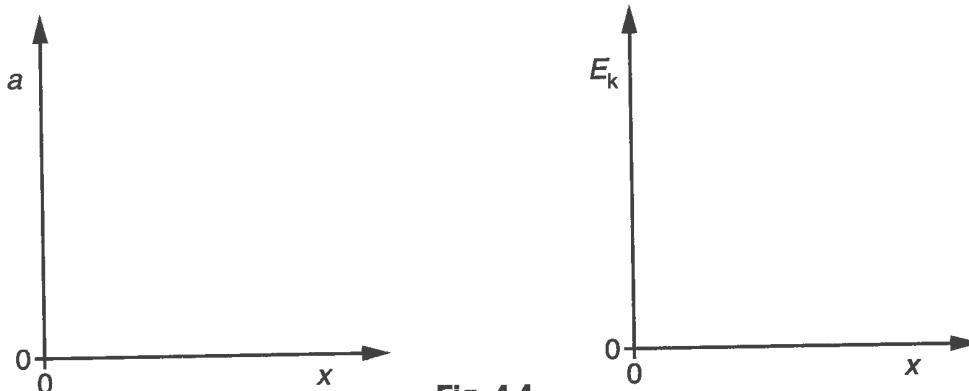
- (i)** Determine the maximum compression of the spring caused by the two balls.

maximum compression =m [3]

- (ii)** On Fig. 4.4, sketch graphs to show the variation with compression x of the spring, from zero to maximum compression, of:

1. the magnitude of the deceleration a of the balls
2. the kinetic energy E_k of the balls.

Numerical values are not required.

**Fig. 4.4**

[3]

41. M/J 19/P22/Q2/c,d

booklet (scan + OCR)

- (a) A short time later, the car in (b) is travelling at a constant speed and the tension in the tow-bar is 480 N.

The tow-bar is a solid metal rod that obeys Hooke's law. Some data for the tow-bar are listed below.

Young modulus of metal = 2.2×10^{11} Pa

original length of tow-bar = 0.48 m

cross-sectional area of tow-bar = 3.0×10^{-4} m²

Determine the extension of the tow-bar.

extension = m [3]

- (b) The driver of the car in (b) sees a pedestrian standing directly ahead in the distance. The driver operates the horn of the car from time $t = 15$ s to $t = 17$ s. The frequency of the sound heard by the pedestrian is 480 Hz. The speed of the sound in the air is 340 m s^{-1} .

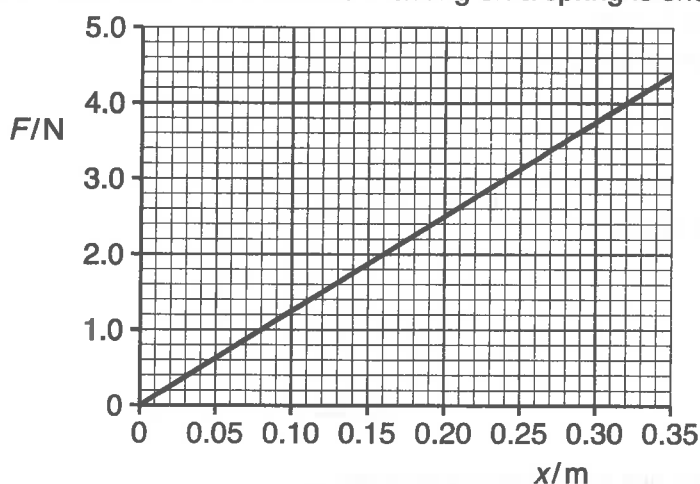
Use Fig. 2.2 to calculate the frequency of the sound emitted by the horn.

frequency = Hz [2]

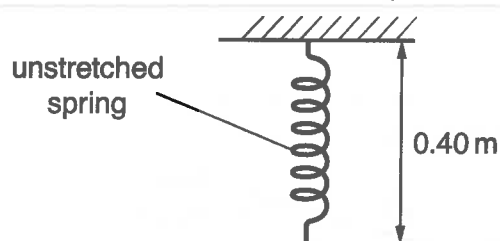
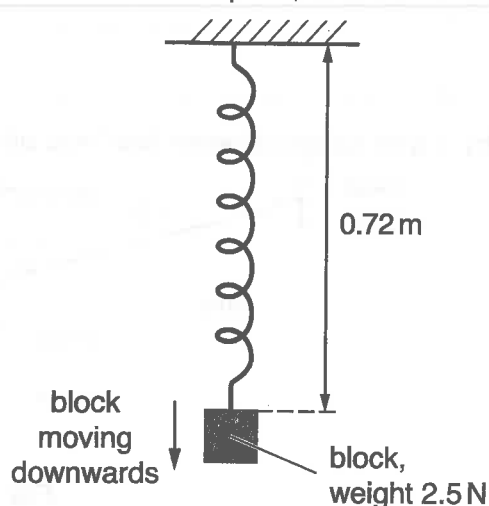
42. M/J 19/P21/Q3

booklet (scan + OCR)

The variation with extension x of the force F acting on a spring is shown in Fig. 3.1.

**Fig. 3.1**

The spring of unstretched length 0.40 m has one end attached to a fixed point, as shown in Fig. 3.2.

**Fig. 3.2****Fig. 3.3**

A block of weight 2.5 N is then attached to the spring. The block is then released and begins to move downwards. At one instant, as the block is continuing to move downwards, the spring has a length of 0.72 m, as shown in Fig. 3.3.

Assume that the air resistance and the mass of the spring are both negligible.

(a) For the change in length of the spring from 0.40 m to 0.72 m:

(i) use Fig. 3.1 to show that the increase in elastic potential energy of the spring is 0.64 J

[2]

(ii) calculate the decrease in gravitational potential energy of the block of weight 2.5 N.

decrease in potential energy = J [2]

(b) Use the information in (a)(i) and your answer in (a)(ii) to determine, for the instant when the length of the spring is 0.72 m:

(i) the kinetic energy of the block

kinetic energy = J [1]

(ii) the speed of the block.

speed = ms^{-1} [2]

43. O/N 18/P23/Q1

booklet selection; question from the official paper

Question from the official paper: printed pages 313–314 of the scanned booklet are missing

- 1 (a) Mass, length and time are all SI base quantities.

State two other SI base quantities.

1.

2.

[2]

- (b) A wire hangs between two fixed points, as shown in Fig. 1.1.

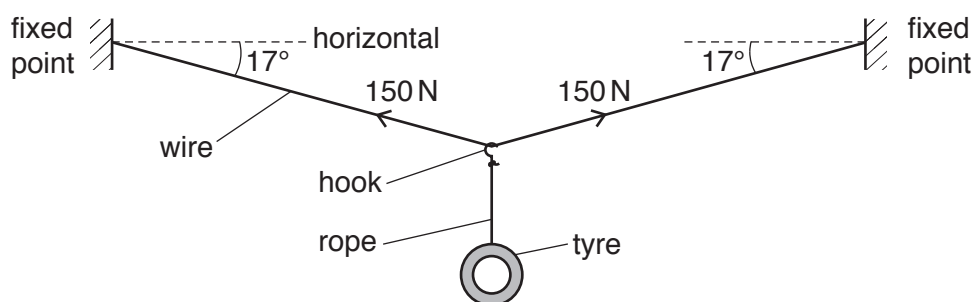


Fig. 1.1 (not to scale)

A child's swing is made by connecting a car tyre to the wire using a rope and a hook. The system is in equilibrium with the wire hanging at an angle of 17° to the horizontal. The tension in the wire is 150 N . Assume that the rope and hook have negligible weight.

- (i) Determine the weight of the tyre.

weight = N [2]

- (ii) The wire has a cross-sectional area of 7.5 mm^2 and is made of metal of Young modulus $2.1 \times 10^{11}\text{ Pa}$. The wire obeys Hooke's law.

Calculate, for the wire,

1. the stress,

stress = Pa [2]

2. the strain.

strain = [2]

44. M/J 18/P21/Q2

booklet selection; question and answer from the official paper

The scanned booklet lost this question (missing pages); only its answer survives

- 2 (a) State Newton's first law of motion.

[1]

- (b) A block of weight 15 N hangs by a wire from a remotely controlled aircraft, as shown in Fig. 2.1.

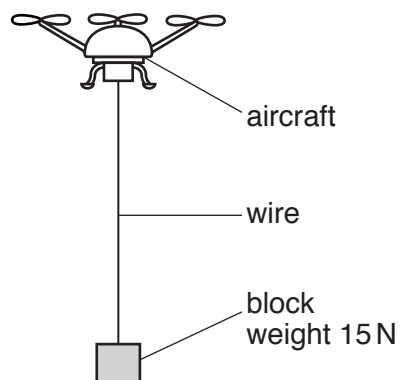


Fig. 2.1

The aircraft is used to move the block only in a vertical direction. The force on the block due to air resistance is negligible.

The variation with time t of the vertical velocity v of the block is shown in Fig. 2.2. The velocity is taken to be positive in the upward direction.

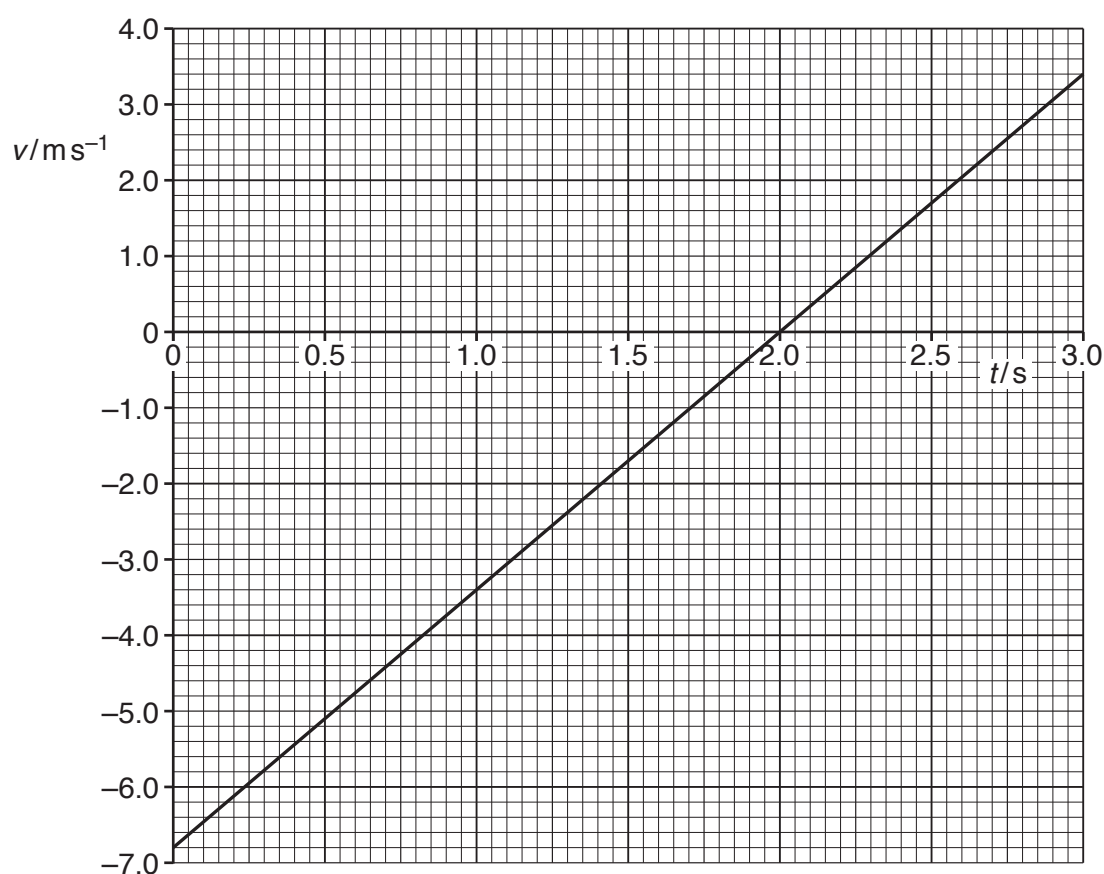


Fig. 2.2

- (i) Determine, for the block,
1. the displacement from time $t = 0$ to $t = 3.0$ s,
magnitude of displacement =

m

direction of displacement

[3]

2. the change in gravitational potential energy from time $t = 0$ to $t = 3.0$ s.

change in gravitational potential energy =

J [2]

(ii) Calculate the magnitude of the acceleration of the block at time $t = 2.0$ s.

acceleration =

m s^{-2} [2]

(iii) Use your answer in (b)(ii) to show that the tension T in the wire at time $t = 2.0$ s is 20 N.

[2]

(iv) The wire has a cross-sectional area of $2.8 \times 10^{-5} \text{ m}^2$ and is made from metal of Young modulus $1.7 \times 10^{11} \text{ Pa}$. The wire obeys Hooke's law.

Calculate the strain of the wire at time $t = 2.0$ s.

strain =

[3]

(v) At some time after $t = 3.0$ s the tension in the wire has a constant value of 15 N.

State and explain whether it is possible to deduce that the block is moving vertically after $t = 3.0$ s.

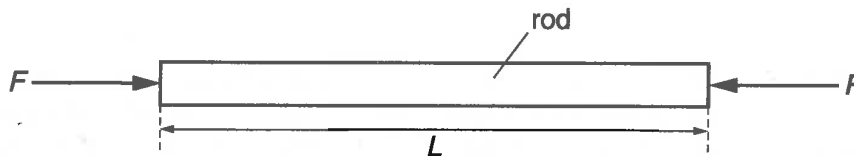
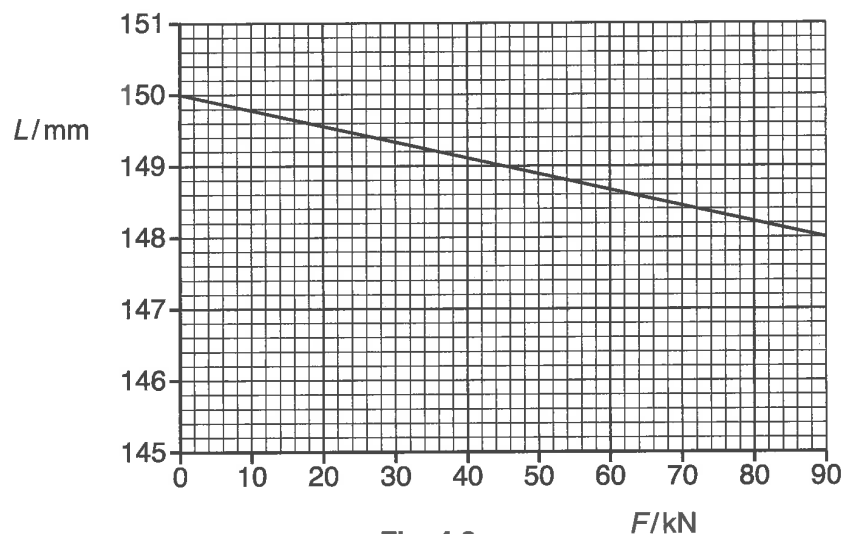
[2]

45. M/J 18/P23/Q4

booklet (scan + OCR)

(a) Define the *Young modulus* of a material.

..... [1]

(b) A metal rod is compressed, as shown in Fig. 4.1.**Fig. 4.1**The variation with compressive force F of the length L of the rod is shown in Fig. 4.2.**Fig. 4.2**

Use Fig. 4.2 to

(i) determine the spring constant k of the rod,

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

(ii) determine the strain energy stored in the rod for $F = 90 \text{ kN}$.

$$\text{strain energy} = \dots\dots\dots \text{J} \quad [3]$$

(c) The rod in **(b)** has cross-sectional area A and is made of metal of Young modulus E . It is now replaced by a new rod of the same original length. The new rod has cross-sectional area $A/3$ and is made of metal of Young modulus $2E$. The compression of the new rod obeys Hooke's law.

On Fig. 4.2, sketch the variation with F of the length L for the new rod from $F = 0$ to $F = 90 \text{ kN}$.

[2]

46. MAR 18/P22/Q3

booklet (scan + OCR)

(a) For the deformation of a wire under tension, define**(i)** *stress*,

..... [1]

(ii) *strain*.

..... [1]

- (b) A wire is fixed at one end so that it hangs vertically. The wire is given an extension x by suspending a load F from its free end. The variation of F with x is shown in Fig. 3.1.

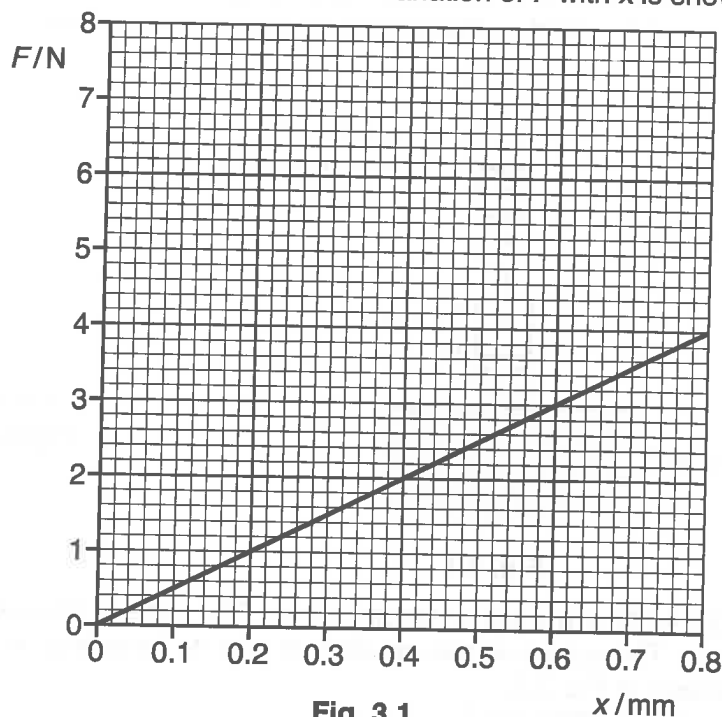


Fig. 3.1

The wire has cross-sectional area $9.4 \times 10^{-8} \text{ m}^2$ and original length 2.5 m.

- (i) Describe how measurements can be taken to determine accurately the cross-sectional area of the wire.

.....

 [3]

- (ii) Determine the Young modulus E of the material of the wire.

$$E = \dots\dots\dots \text{ Pa [2]}$$

- (iii) Use Fig. 3.1 to calculate the increase in the energy stored in the wire when the load is increased from 2.0 N to 4.0 N.

$$\text{increase in energy} = \dots\dots\dots \text{ J [2]}$$

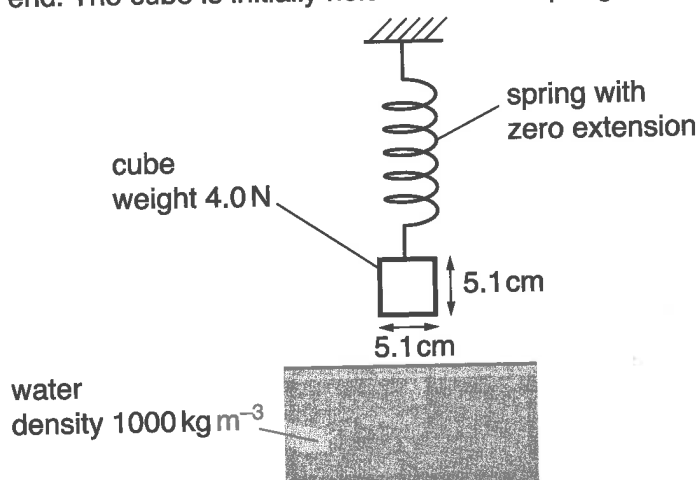
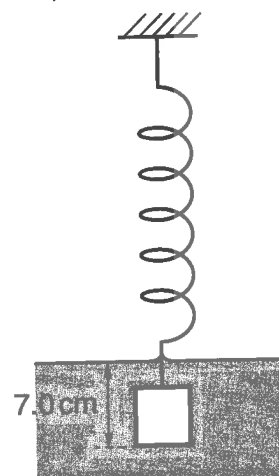
- (c) The wire in (b) is replaced by a new wire of the same material. The new wire has twice the length and twice the diameter of the old wire. The new wire also obeys Hooke's law.

On Fig. 3.1, sketch the variation with extension x of the load F for the new wire from $x = 0$ to $x = 0.80 \text{ mm}$. [2]

47. O/N 17/P22/Q3

booklet (scan + OCR)

A spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially held so that the spring has zero extension, as shown in Fig. 3.1.

**Fig. 3.1****Fig. 3.2**

The cube has weight 4.0 N and sides of length 5.1 cm. The cube is released and sinks into water as the spring extends. The cube reaches equilibrium with its base at a depth of 7.0 cm below the water surface, as shown in Fig. 3.2.

The density of the water is 1000 kg m⁻³.

- (a) Calculate the difference in the pressure exerted by the water on the bottom face and on the top face of the cube.

difference in pressure = Pa [2]

- (b) Use your answer in (a) to show that the upthrust on the cube is 1.3 N.

[2]

- (c) Calculate the force exerted on the spring by the cube when it is in equilibrium in the water.

force = N [1]

- (d) The spring obeys Hooke's law and has a spring constant of 30 N m⁻¹. Determine the initial height above the water surface of the base of the cube before it was released.

height above surface = cm [3]

(e) The cube in the water is released from the spring.

(i) Determine the initial acceleration of the cube.

acceleration = ms^{-2} [2]

(ii) Describe and explain the variation, if any, of the acceleration of the cube as it sinks in the water.

.....
 [2]

48. O/N 17/P21/Q4

booklet (scan + OCR)

(a) Define *strain*.

..... [1]

(b) A wire is designed to ensure that its strain does not exceed 4.0×10^{-4} when a force of 8.0 kN is applied. The Young modulus of the metal of the wire is 2.1×10^{11} Pa. It may be assumed that the wire obeys Hooke's law.

For a force of 8.0 kN, calculate, for the wire,

(i) the maximum stress,

maximum stress = Pa [2]

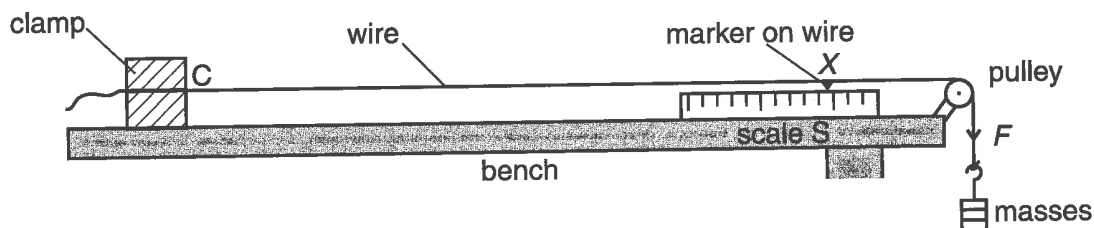
(ii) the minimum cross-sectional area.

minimum cross-sectional area = m^2 [2]

49. M/J 17/P22/Q3

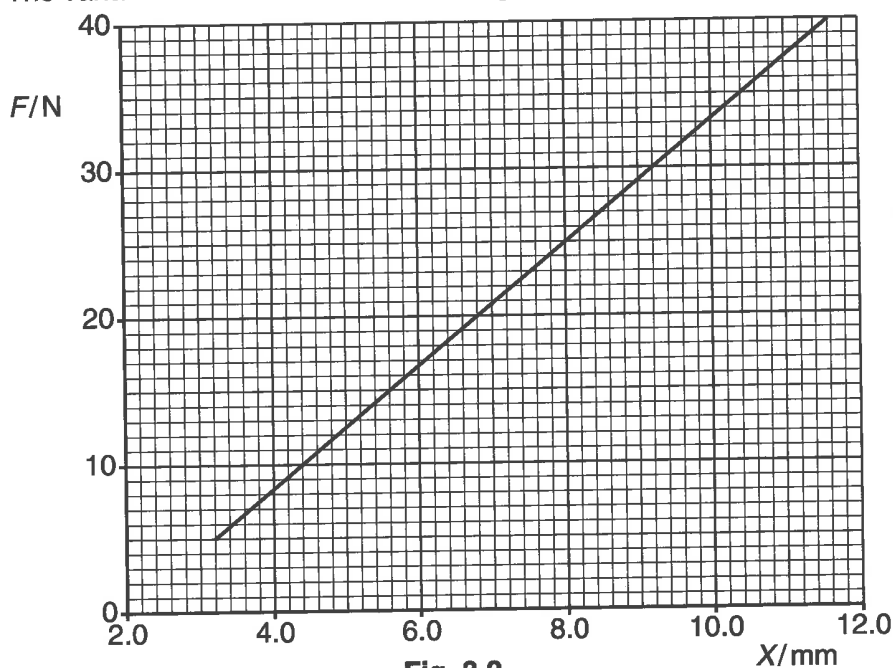
booklet (scan + OCR)

The Young modulus of the material of a wire can be determined using the apparatus shown in Fig. 3.1.

**Fig. 3.1**

One end of the wire is clamped at C and a marker is attached to the wire above a scale S. A force to extend the wire is applied by attaching masses to the other end of the wire.

The reading X of the marker on the scale S is determined for different forces F applied to the end of the wire. The variation with X of F is shown in Fig. 3.2.

**Fig. 3.2**

- (a) The length of the wire from C to the marker for $F = 0$ is 3.50 m. The diameter of the wire is 0.38 mm.

Use the gradient of the line in Fig. 3.2 to determine the Young modulus E of the material of the wire in TPa.

$$E = \dots\dots\dots \text{TPa} \quad [3]$$

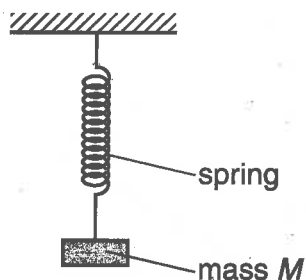
- (b) The experiment is repeated with a thicker wire of the same material and length.
State how the range of the force F must be changed to obtain the same range of scale readings as in Fig. 3.2.

..... [1]

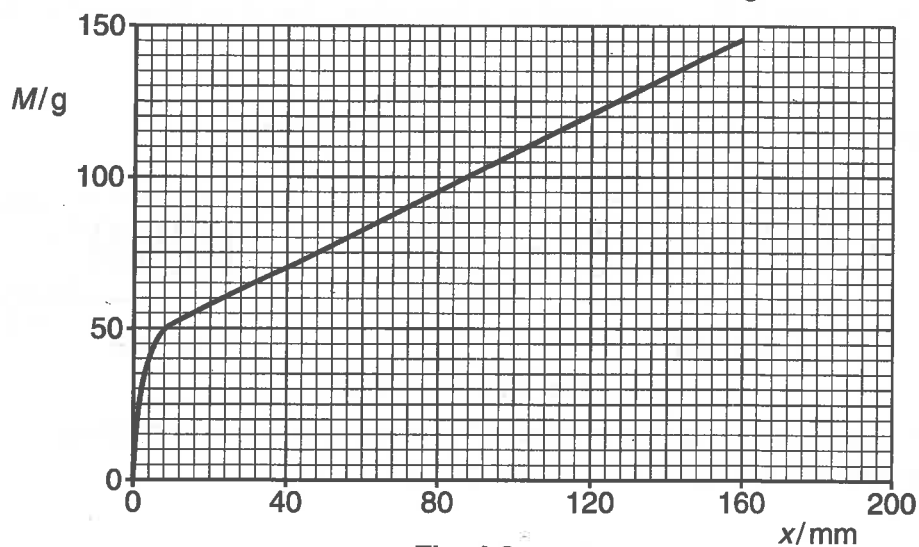
50. M/J 17/P23/Q4

booklet (scan + OCR)

A spring is supported so that it hangs vertically, as shown in Fig. 4.1.

**Fig. 4.1**

Different masses are attached to the lower end of the spring. The extension x of the spring is measured for each mass M . The variation with x of M is shown in Fig. 4.2.

**Fig. 4.2**

(a) State and explain whether the spring obeys Hooke's law.

..... [1]

(b) State the form of energy stored in the spring due to the addition of the masses.

..... [1]

(c) Describe how to determine whether the extension of the spring is elastic.

..... [1]

(d) Calculate the work done on the spring as it is extended from $x = 40.0$ mm to $x = 160$ mm.

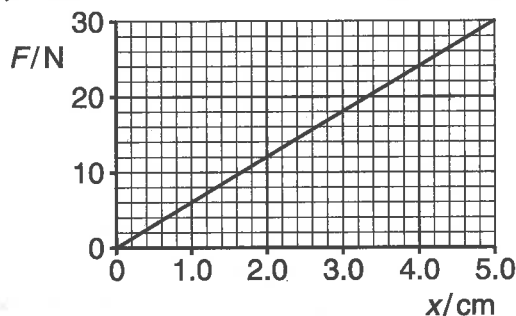
work done = J [3]

51. O/N 16/P21/Q3

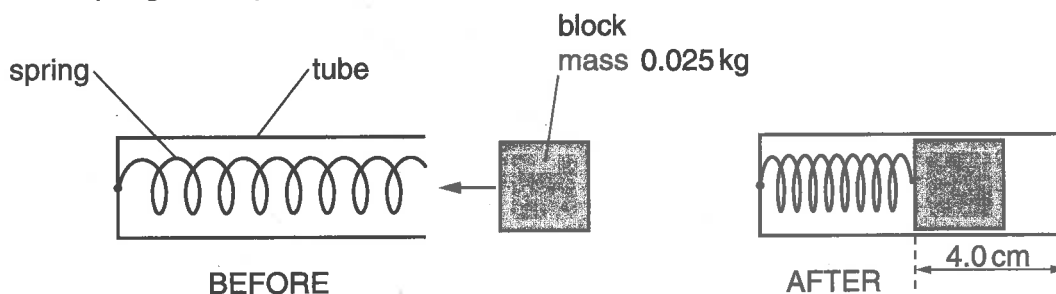
booklet (scan + OCR)

(a) State Hooke's law.

[1]

(b) The variation with compression x of the force F acting on a spring is shown in Fig. 3.1.**Fig. 3.1**

The spring is fixed to the closed end of a horizontal tube. A block is pushed into the tube so that the spring is compressed, as shown in Fig. 3.2.

**Fig. 3.2** (not to scale)

The compression of the spring is 4.0 cm. The mass of the block is 0.025 kg.

(i) Calculate the spring constant of the spring.spring constant = N m^{-1} [2]**(ii)** Show that the work done to compress the spring by 4.0 cm is 0.48 J.

[2]

(iii) The block is now released and accelerates along the tube as the spring returns to its original length. The block leaves the end of the tube with a speed of 6.0 m s^{-1} .

1. Calculate the kinetic energy of the block as it leaves the end of the tube.

kinetic energy = J [2]

2. Assume that the spring has negligible kinetic energy as the block leaves the tube. Determine the average resistive force acting against the block as it moves along the tube.

resistive force = N [3]

- (iv) Determine the efficiency of the transfer of elastic potential energy from the spring to the kinetic energy of the block.

efficiency = [2]

52. M/J 16/P22/Q3

booklet (scan + OCR)

- (a) Define the *Young modulus*.

..... [1]

- (b) The Young modulus of steel is 1.9×10^{11} Pa. The Young modulus of copper is 1.2×10^{11} Pa. A steel wire and a copper wire each have the same cross-sectional area and length. The two wires are each extended by equal forces.

- (i) Use the definition of the Young modulus to determine the ratio

$$\frac{\text{extension of the copper wire}}{\text{extension of the steel wire}}$$

ratio = [3]

- (ii) The two wires are each extended by a force. Both wires obey Hooke's law. On Fig. 3.1, sketch a graph for each wire to show the variation with extension of the force. Label the line for steel with the letter **S** and the line for copper with the letter **C**.

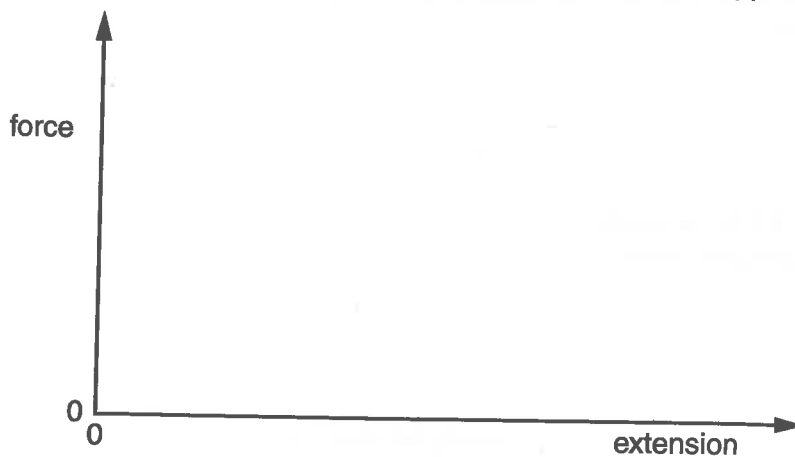


Fig. 3.1

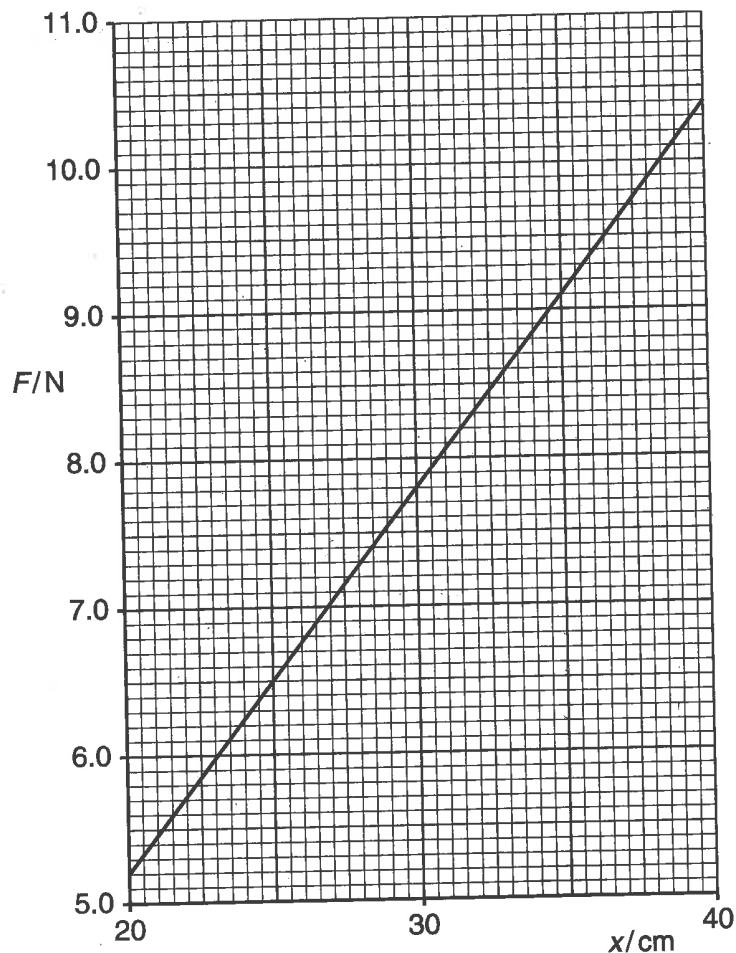
[1]

53. M/J 16/P21/Q4

booklet (scan + OCR)

(a) State what is meant by *elastic potential energy*.

[1]

(b) A spring is extended by applying a force. The variation with extension x of the force F is shown in Fig. 4.1 for the range of values of x from 20 cm to 40 cm.**Fig. 4.1****(i)** Use data from Fig. 4.1 to show that the spring obeys Hooke's law for this range of extensions.

[2]

(ii) Use Fig. 4.1 to calculate
1. the spring constant,spring constant = Nm^{-1} [2]2. the work done extending the spring from $x = 20\text{ cm}$ to $x = 40\text{ cm}$.

work done = J [3]

(c) A force is applied to the spring in **(b)** to give an extension of 50 cm.

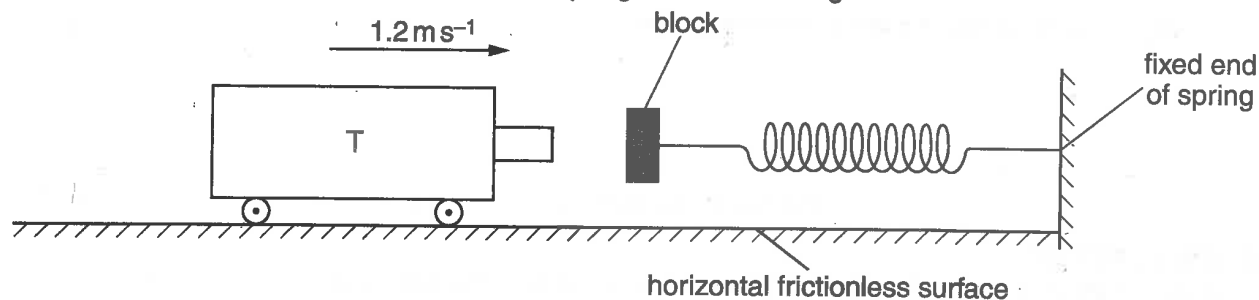
State how you would check that the spring has not exceeded its elastic limit.

[1]

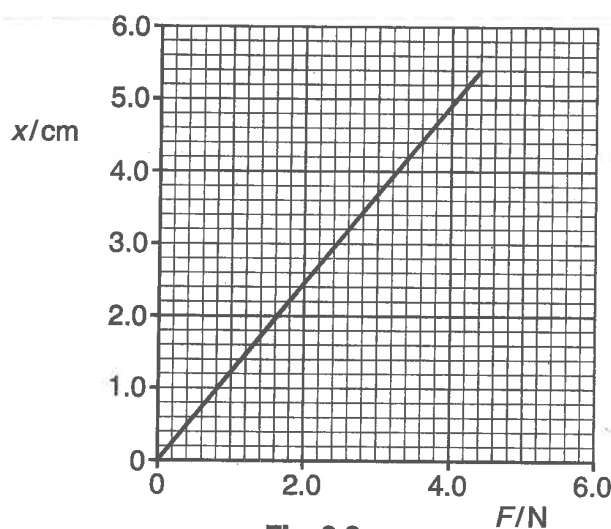
54. O/N 15/P22/Q3

booklet (scan + OCR)

A trolley T moves at speed 1.2 ms^{-1} along a horizontal frictionless surface. The trolley collides with a stationary block on the end of a fixed spring, as shown in Fig. 3.1.

**Fig. 3.1**

The mass of T is 250 g. T compresses the spring by 5.4 cm as it comes to rest. The relationship between the force F applied to the block and the compression x of the spring is shown in Fig. 3.2.

**Fig. 3.2**

- (a) Use Fig. 3.2 to determine
 (i) the spring constant of the spring,

spring constant = Nm^{-1} [2]

- (ii) the work done by T compressing the spring by 5.4 cm.

work done = J [2]

- (b) The spring then expands and causes T to move in a direction opposite to its initial direction. At the time that T loses contact with the block, it is moving at a speed of 0.75 m s^{-1} . From the time that T is in contact with the block,

- (i) describe the energy changes,

.....
 [2]

- (ii) determine the change in momentum of T.

change in momentum = N s [2]

55. O/N 15/P23/Q7

booklet (scan + OCR)

A steel wire of cross-sectional area 15 mm^2 has an ultimate tensile stress of $4.5 \times 10^8 \text{ N m}^{-2}$.

- (a) Calculate the maximum tension that can be applied to the wire.

tension = N [2]

- (b) The steel of the wire has density 7800 kg m^{-3} . The wire is hung vertically. Calculate the maximum length of the steel wire that could be hung vertically before the wire breaks under its own weight.

length = m [3]

Answers Section

1. M/J 26/P22/Q2/c

2(c)(i)	Hooke's (law)	A1	Allow phonetic misspellings of 'Hooke' e.g. Hook, Huk, Hookie
2(c)(ii)	When the tension is removed, the ropes have original length / extension is zero OR when the tension is removed object is not deformed	A1	Allow 'it' to mean ropes Allow 'force' / 'load' / 'T' / 'stress' / weight' for tension Ignore 'mass' for tension Allow 'unloaded' for tension is removed Allow 'tension is released' (BOD) but ignore 'object released' Allow 'unstretched' / 'initial' for original Allow 'shape <u>and</u> size' for length Allow 'dimensions' for length Ignore 'state' / 'form' for length

2. M/J 26/P23/Q2/a,b(ii,iii),c,d

2(a)	stress per unit strain	B1	Allow stress per (unit) strain Allow stress divided by strain Allow stress over strain Allow stress/strain Allow <u>ratio</u> of stress by strain (BOD) Allow ratio of stress and/to strain Ignore stress by strain or stress into strain Ignore symbols unless defined Ignore reference to 'object' or 'material' etc. Ignore any units Ignore 'stiffness'
2(b)(ii)	$(\sigma =) F / A$ $= (0.018 \times 9.81) / (2.6 \times 10^{-6}) = 6.8 \times 10^4 \text{ (Pa)}$	C1 A1	 Full substitution and answer needed. not $g = 10$ as leads to 6.9×10^4
2(b)(iii)	weight (of thread supported at a point) decreases as distance (from top of thread) increases (uniform cross-section so) stress (at a point in the thread) is proportional to the weight (of thread below the point)	B1 B1	Allow weight depends on distance / length Allow weight varies with distance / length Not weight is <u>proportional</u> to distance Not stress is proportional to distance Allow 'force'/'mass' instead of 'weight' throughout
2(c)(i)	$\epsilon = x / L$ or $E = \sigma / \epsilon$ or $x = \sigma L / E$ extension $= 7.2 \times (6.8 \times 10^4) / (8.0 \times 10^8)$ $= 6.1 \times 10^{-4} \text{ m}$	C1 A1	Allow e or ΔL for x
2(c)(ii)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}F^2 / k$ $= \frac{1}{2} \times 6.8 \times 10^4 \times 2.6 \times 10^{-6} \times 6.1 \times 10^{-4}$ or $= \frac{1}{2} \times 0.018 \times 9.81 \times 6.1 \times 10^{-4}$ or $= \frac{1}{2} \times (6.8 \times 10^4 \times 2.6 \times 10^{-6} / 6.1 \times 10^{-4}) \times (6.1 \times 10^{-4})^2$ or $= \frac{1}{2} \times (0.018 \times 9.81 / 6.1 \times 10^{-4}) \times (6.1 \times 10^{-4})^2$ or $= \frac{1}{2} \times (6.8 \times 10^4 \times 2.6 \times 10^{-6})^2 / (6.8 \times 10^4 \times 2.6 \times 10^{-6} / 6.1 \times 10^{-4})$ or $= \frac{1}{2} \times (0.018 \times 9.81)^2 / (0.018 \times 9.81 / 6.1 \times 10^{-4})$ $= 5.4 \times 10^{-5} \text{ J}$	C1 A1	Allow W or E_p or EPE for E ECF x from (c)(i) . $k = 6.8 \times 10^4 \times 2.6 \times 10^{-6} / 6.1 \times 10^{-4} = 290$
2(d)	(extension in 2(b) is) less (than in 2(c)(i)) because stress (in vertical thread) is less (than $6.8 \times 10^4 \text{ Pa}$ everywhere except the top)	B1	Allow 'force' for 'stress' ignore 'weight' / 'mass' for stress

3. M/J 26/P24/Q2/a,c

2(a)	Extension is proportional to (applied) force	B1	Ignore symbols unless defined. allow 'force is proportional to extension' Allow force $\propto$ extension allow load / tension for force ignore reference to the elastic limit ignore force = spring constant $\times$ extension
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2(c)(i)	The extension will be less for Y as the spring constant is greater (and the applied force is the same).	B1	or vice-versa Assume refers to Y unless otherwise stated Allow syllabus symbols not defined, $x/k/F$ Allow the extension is less as extension is inversely proportional to spring constant Allow $x \downarrow$ and $k \uparrow$
2(c)(ii)	(elastic potential energy) will be less for Y	B1	or vice-versa. Assume refers to Y unless stated otherwise Allow $EPE/E/\Delta E/E_P$
	As the extension is less (for the same force) and $E_P = (\frac{1}{2}Fx)/(\frac{1}{2}mgx)$	B1	Allow syllabus symbols $x/k/F$ not defined
	OR As the spring constant is greater (for the same force) and $E_P = \frac{1}{2}F^2/k$	(B1)	Allow $k \uparrow$ Allow $E_P = \frac{1}{2}(mg)^2/k$ ignore $E = \frac{1}{2}kx^2$

4. MAR 26/P22/Q4/a(iv)

4(a)(iv)	$E = \sigma/\epsilon$ or $E = FL/Ax$ and $\epsilon = x/L$	C1
	$\epsilon = F/(AE)$	
	$= 13/(\pi(0.8 \times 10^{-3}/2)^2 \times 9.3 \times 10^{10})$	C1
	$= 2.8 \times 10^{-4}$	A1

5. O/N 25/P21/Q3/a,b(ii),c(ii),d(ii)

3(a)	ratio of stress to strain	B1
3(b)(ii)	$k = F/x$	C1
	$E = FL/Ax = ((F/x) \times (L/A)) = kL/A$ leading to $k_0 = EA/L$	A1
3(c)(ii)	straight horizontal line starting at (0, k_0)	B1
3(d)(ii)	$k = [1.3 \times 10^{11} \times \pi \times (0.80 \times 10^{-3})^2] / 3.8$	C1
	or $k = EA/[RA/\rho] = E\rho/R$ $= [1.3 \times 10^{11} \times 1.8 \times 10^{-8}] / 0.034$	
	$k = 6.9 \times 10^4 \text{ N m}^{-1}$	A1

6. O/N 25/P22/Q3/b(ii),c

3(b)(ii)	$x = F/k$	C1
	$= 110/250$	A1
	$= 0.44 \text{ m}$	
3(c)(i)	extension = $55/250$ or $110/(2 \times 250)$	C1
	extension = 0.22 m	A1

3(c)(ii)	$E = \frac{1}{2} Fx$ or $E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} F^2 / k$	C1
	$E_1 / E_2 = (\frac{1}{2} \times 110 \times 0.44) / (2 \times (\frac{1}{2} \times 55 \times 10.22))$ or $E_1 / E_2 = (\frac{1}{2} k \times 0.44^2) / (2 \times (\frac{1}{2} k \times 0.22^2))$ or $E_1 / E_2 = (\frac{1}{2} \times 110^2 / k) / (2 \times (\frac{1}{2} \times 55^2 / k))$ $E_1 / E_2 = 2.0$ (no ECF from 3(b)(ii) and 3(c)(i))	A1

7. O/N 25/P24/Q3/a,b,c(ii)

3(a)	force is proportional to extension	B1
3(b)(i)	length = 37 m	A1
3(b)(ii)	spring constant = F / x	C1
	= e.g. $1800 / (120 - 37)$ = 22 N m^{-1}	A1
3(c)(ii)	$(E =) \frac{1}{2} Fx$ or $(E =) \frac{1}{2} kx^2$ or $(E =) \frac{1}{2} F^2 / k$ or $(E =)$ area under graph	C1
	$(E =) \frac{1}{2} \times 1800 \times (120 - 37) = 75\,000 \text{ J}$ or 75 kJ or $(E =) \frac{1}{2} \times 21.7 \times (120 - 37)^2 = 75\,000 \text{ J}$ or 75 kJ	A1

8. M/J 25/P21/Q2/c

2(c)(i)	$k = F / x$	C1
	$x = 0.80 - 0.59$ = 0.21 m	C1
	$k = (60 \times 9.81) / 0.21$ = 2800 N m^{-1}	A1
2(c)(ii)	$E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} Fx$ or $E = \frac{1}{2} F^2 / k$ = $\frac{1}{2} \times 2800 \times 0.21^2$ or = $\frac{1}{2} \times 60 \times 9.81 \times 0.21$ or = $\frac{1}{2} \times (60 \times 9.81)^2 / 2800$	C1
	= 62 J	A1

9. M/J 25/P22/Q5/a,b

5(a)	the ratio of stress to strain	B1
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5(b)(i)	$A = FL / Ex$	C1
	$A = \text{e.g. } (500 \times 0.81) / (95 \times 10^9 \times 4.0 \times 10^{-3})$	C1
	$A = 1.1 \times 10^{-6} \text{ m}^2$	A1
	OR $A = kL / E$ or $A = \text{gradient} \times L / E$ $k = \text{e.g. } 500 / 4.0 \times 10^{-3}$ $k = 1.25 \times 10^5$	(C1)
	$A = 1.25 \times 10^5 \times 0.81 / 95 \times 10^9$	(C1)
	$A = 1.1 \times 10^{-6} \text{ m}^2$	(A1)
5(b)(ii)	$E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} Fx$ or $E = \text{area (under graph)}$	C1
	$(\Delta)E = \frac{1}{2} \times 1.25 \times 10^5 \times ((3.0 \times 10^{-3})^2 - (2.0 \times 10^{-3})^2)$ or $(\Delta)E = (\frac{1}{2} \times 375 \times 3.0 \times 10^{-3}) - (\frac{1}{2} \times 250 \times 2.0 \times 10^{-3})$ or $(\Delta)E = \frac{1}{2} \times (375 + 250) \times 1.0 \times 10^{-3}$	C1
	work done = 0.31 J	A1

10. M/J 25/P23/Q2/b(ii,iii)

2(b)(ii)	Young modulus = σ / ϵ or $F / A\epsilon$ or FL / Ax	C1
	Area of wire = $\pi \times (1.8 \times 10^{-3} / 2)^2$ = 2.5×10^{-6}	C1
	So Young modulus = $((90 \times 9.81) / \pi \times (9.0 \times 10^{-4})^2) / 1.2 \times 10^{-3}$ = $2.9 \times 10^{11} \text{ Pa}$	A1
2(b)(iii)	Moment provided by B will decrease / moment due to wire will increase	B1
	So force acting on wire will increase (Young's modulus and area remain constant) and the strain will increase	B1

11. M/J 25/P24/Q3/d

3(b)	$(\Delta)E_{(P)} / 20 = mg(\Delta)h$ OR $(\Delta)E_{(P)} / 20 = mgx_0$	C1
	$(x_0 =) 20 / 5.5 \times 9.81 = 0.37 \text{ (m)}$ Allow $(\Delta)h$ for x_0	A1
3(c)	$[\max (E_{(P)})] = 110 + 20 = 130 \text{ J}$	A1
3(d)	$(\max) E_{(P)} / 130 = \frac{1}{2} F_{(0)} x_{(0)}$	C1
	$(F_0 =) 2 \times 130 / 0.37 = 700 \text{ (N)}$	A1
	or $(\max) E_{(P)} / 130 = 1 / 2k x_{(0)}^2$ $(F_0 =) k x_{(0)}$	(C1)
	$(F_0 =) 2 \times 130 \times 0.37 / (0.37)^2 = 700 \text{ (N)}$	(A1)

12. O/N 24/P21/Q4/a,b,c

4(a)	stress per unit strain	B1
4(b)(i)	straight line with positive gradient passing through origin	B1
4(b)(ii)	spring constant	B1
4(b)(iii)	elastic potential energy (stored in wire)	B1

4(c)	length (of Q) is half (the length of P)	B1
	extension is proportional to length, and inversely proportional to area and Young modulus	B1
	extension of Q is $= \frac{1}{2} / (2 \times 2)$ $= \frac{1}{8}$ times the extension of P	B1

13. O/N 24/P22/Q3/a,c(ii,iii)

3(a)(i)	$E = \sigma / \epsilon$ or $E = \text{gradient}$	C1
	$E = \text{e.g. } 12 \times 10^7 / 0.0050$ $= 2.4 \times 10^{10} \text{ Pa}$	A1
3(a)(ii)	cross drawn at (1.0%, $24 \times 10^7 \text{ Pa}$), labelled Q	B1
3(c)(ii)	$\sigma = F / A$	C1
	$\pi r^2 = 46 / (1.5 \times 10^7)$ $r = 9.9 \times 10^{-4} \text{ m}$	A1
3(c)(iii)	elastic limit is not reached or (new) stress is less than (stress at) elastic limit or (new) strain is less than (strain at) elastic limit	M1
	(so the wire behaves) elastically	A1

14. O/N 24/P23/Q4/a,b

4(a)(i)	(normal) force per unit cross-sectional area	B1
4(a)(ii)	extension per unit unstretched length	B1
4(b)(i)	$E = FL / Ax$	C1
	$A = (9.0 \times 0.84) / (1.9 \times 10^9 \times 0.47 \times 10^{-3})$	C1
	$= 8.5 \times 10^{-6} \text{ m}^2$	A1
4(b)(ii)	F , L and x are all the same (for both wires / as in X) or F and strain are the same	B1
	A is greater (for Y), so the Young modulus (for Y) is less than $1.9 \times 10^9 \text{ Pa}$ or less than that of wire X	B1

15. M/J 24/P21/Q4/a,b,c

4(a)	extension / <u>original</u> length				B1
4(b)(i)	Young modulus = stress / strain				C1
	$= (18 / 4.5 \times 10^{-7}) / (1.4 \times 10^{-3} / 4.0)$				C1
	$= (4.0 \times 10^7) / (3.5 \times 10^{-4})$				A1
	$= 1.1 \times 10^{11} \text{ Pa}$				
4(b)(ii)	straight line through the origin				B1
	ending at the point (3.5, 4.0)				B1
4(c)		greater in second wire	less in second wire	the same in both wires	B2
	stress		✓		
	strain		✓		

16. M/J 24/P22/Q3/d

3(d)	$E = \sigma / \epsilon$	C1
	$x = \sigma L / E$	C1
	$= 1.9 \times 10^6 \times 0.12 / (1.3 \times 10^{11})$ $= 1.8 \times 10^{-6} \text{ m}$	A1

17. M/J 24/P22/Q4/a

4(a)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
	$E = \frac{1}{2} \times 29 \times (8.0 \times 10^{-2})^2$ or $E = \frac{1}{2} \times 2.32 \times 8.0 \times 10^{-2}$ $E = 9.3 \times 10^{-2} \text{ J}$	A1

18. M/J 24/P23/Q3/a,b,c,d,e

3(a)	extension is proportional to (applied) force	B1
3(b)(i)	P at (60, 5.4)	A1
3(b)(ii)	E at (80, 5.9)	A1
3(c)(i)	$k = F / x$ or $k = \text{gradient of (straight line section of) graph}$	C1
	e.g. gradient = $5.4 / 0.060$ $k = 90 \text{ N m}^{-1}$	A1
3(c)(ii)	Young modulus or $E = \sigma / \epsilon$ or FL / Ax or kL / A	C1
	$E = (5.4 \times 3.2) / (4.0 \times 10^{-7} \times 0.06)$ or $90 \times 3.2 / (4.0 \times 10^{-7})$	C1
	$E = 7.2 \times 10^8 \text{ Pa}$	A1
3(d)	work done = area under graph	B1
	$= (1.0 \pm 0.2) \text{ J}$	A1
3(e)	the extension will be smaller (for the same force on the thicker sample) or a greater force is required (to extend the thicker sample by the same amount) or spring constant is proportional to area	M1
	the spring constant (of the second sample) will be greater	A1

19. MAR 24/P22/Q3/a,b

3(a)(i)	$\sigma = 0.72 \times 10^9$	C1
	force = $\sigma \times A$ $= 0.72 \times 10^9 \times \pi \times (1.2 \times 10^{-3} / 2)^2$	C1
	$= 810 \text{ N}$	A1
	or Young modulus = gradient of graph e.g. $= 0.80 \times 10^9 / 6.0 \times 10^{-3}$ $= 1.33 \times 10^{11}$	(C1)
	force = Young modulus $\times$ strain $\times A$ $= 1.33 \times 10^{11} \times 5.4 \times 10^{-3} \times \pi \times (1.2 \times 10^{-3} / 2)^2$	(C1)
	$= 810 \text{ N}$	(A1)
3(a)(ii)	$E_{(P)} = \frac{1}{2} Fx$ or $E_{(P)} = \frac{1}{2} kx^2$ and $F = kx$	C1
	$x = 2E_P / F$ $x = 2 \times 0.31 / 810$ $x = 7.7 \times 10^{-4}$	C1
	$L = x / \epsilon$ $L = 7.7 \times 10^{-4} / 5.4 \times 10^{-3}$ $L = 0.14 \text{ m}$	A1
	or $E_{(P)} = \frac{1}{2} Fx$ or $E_{(P)} = \frac{1}{2} kx^2$ and $k = EA/L$	(C1)
	$x = 2E_P / EA \epsilon$ $x = 2 \times 0.31 / (1.33 \times 10^{11} \times \pi \times (1.2 \times 10^{-3} / 2)^2 \times 5.4 \times 10^{-3})$ $x = 7.6 \times 10^{-4}$	(C1)
	$L = x / \epsilon$ $L = 7.6 \times 10^{-4} / 5.4 \times 10^{-3}$ $L = 0.14 \text{ m}$	(A1)
3(b)	A straight line, passing through the origin with a larger gradient than wire X.	M1
	Gradient of the line is twice the gradient of wire X.	A1

20. O/N 23/P21/Q2/c

2(c)(i)	(extension $\Rightarrow$) $2.5 \times 2.4 \times 10^{-5} = 6.0 \times 10^{-5} \text{ (m)}$	A1
2(c)(ii)	$E_{(P)} = \frac{1}{2} Fx$ or $E_{(P)} = \frac{1}{2} kx^2$ and $F = kx$	C1
	$E_{(P)} = \frac{1}{2} \times 4.00 \times 10^2 \times 6.0 \times 10^{-5}$ or $E_{(P)} = \frac{1}{2} \times 6.7 \times 10^6 \times (6.0 \times 10^{-5})^2$ $E_{(P)} = 0.012 \text{ J}$	A1
2(c)(iii)	longer extension or smaller spring constant	M1
	elastic potential energy is greater	A1

21. O/N 23/P22/Q3/a,d

3(a)	$E_{(P)} = \frac{1}{2}kx^2$ or $E_{(P)} = \frac{1}{2}Fx$ and $F = kx$	C1
	$0.048 = \frac{1}{2} \times k \times (2.1 \times 10^{-2})^2$ $k = 220 \text{ N m}^{-1}$	A1
3(d)	sketch: curved line from the origin	M1
	curved line has increasing gradient	A1

22. O/N 23/P23/Q4/a,b

4(a)(i)	spring constant	B1
4(a)(ii)	area represents the work done (to extend the wire)	B1
	work done (to extend the wire) is equal to elastic potential energy	B1
4(b)(i)	$x_G = 0.39 \text{ mm}$ and $x_H = 0.29 \text{ mm}$	A1
4(b)(ii)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ and $F = kx$ or $E = \text{area under graph}$	C1
	for G: $E = \frac{1}{2} \times 2.0 \times (0.39 \times 10^{-3})$ or $\frac{1}{2} \times 5.1 \times 10^3 \times (0.39 \times 10^{-3})^2$ for H: $E = \frac{1}{2} \times 2.0 \times (0.29 \times 10^{-3})$ or $\frac{1}{2} \times 6.9 \times 10^3 \times (0.29 \times 10^{-3})^2$ $E_P = 3.9 \times 10^{-4} + 2.9 \times 10^{-4}$ $= 6.8 \times 10^{-4} \text{ J}$	A1
4(b)(iii)	$E = FL / Ax$ or stress / strain = FL / Ax	C1
	$A_G / A_H = 1 \times (0.29 \times 10^{-3}) / [1.5 \times (0.39 \times 10^{-3})]$	C1
	ratio = 0.50	A1

23. M/J 23/P22/Q4**(a)** Hooke's (law)

[1]

(b) $k = F / x$ **or** $k = \text{gradient}$

= e.g. $12.0 / (0.240 - 0.08)$

= 75 N m^{-1}

[2]

(c) $E = \frac{1}{2}Fx$ **or** $E = \frac{1}{2}kx^2$ **or** $E = \text{area under graph}$

$$E = \frac{1}{2} \times 6.0 \times 0.080 \text{ or } \frac{1}{2} \times 75 \times 0.08^2$$

= 0.24 J

[2]

24. M/J 23/P21/Q2/a,b

(a) (i) (component =) $4.8 \sin 30^\circ = 2.4 \text{ (N)}$ [1]

(ii) (8.2×0.50) or $(W \times 0.60)$ or (0.30×0.80) or (2.4×1.2)
 $(8.2 \times 0.50) = (W \times 0.60) + (0.30 \times 0.80) + (2.4 \times 1.2)$

$W = 1.6 \text{ N}$ [3]

(iii) force = $4.8 \cos 30^\circ$
 $= 4.2 \text{ N}$ [1]

(b) $E = \frac{1}{2}Fx$

$0.32 = \frac{1}{2} \times 8.2 \times x$

$x = 0.078 \text{ m}$ [2]

25. M/J 23/P23/Q6

(a) (Young modulus =) stress / strain [1]

(b) (i) unstretched length = 1.9980 m [1]

(ii) stress = F / A
 $= 30 / 9.5 \times 10^{-7}$
 $= 3.2 \times 10^7 \text{ Pa}$

strain = $0.0050 / 1.9980$
 $= 2.5 \times 10^{-3}$ [3]

26. O/N 22/P22/Q4

Answer from the official mark scheme: printed pages 328–329 of the scanned booklet are missing

4(a)	$E = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 0.25 \times 2.3^2$	A1
	$= 0.66 \text{ J}$	
4(b)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
	$x = [(2 \times 0.66) / 420]^{0.5}$ $= 0.056 \text{ m}$	A1
	or	
	$E = \frac{1}{2}kx^2$	(C1)
	$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$ $x = (0.25 \times 2.3^2 / 420)^{0.5}$ $= 0.056 \text{ m}$	(A1)
4(c)(i)	$(p =) 0.25 \times 2.3$ or 0.25×1.5	C1
	change in momentum = $0.25 (2.3 + 1.5)$ $= 0.95 \text{ N s}$	A1
4(c)(ii)	resultant force = $0.95 / 0.086$ or $0.25 \times (2.3 + 1.5) / 0.086$ $= 11 \text{ N}$	A1
4(d)	curved line from the origin with an increasing gradient	B1

27. O/N 22/P21/Q2

Answer from the official mark scheme: printed pages 328–329 of the scanned booklet are missing

2(a)(i)	$t = 1.8 / 4.9$ $= 0.37 \text{ s}$	A1
2(a)(ii)	$v = u + at$ $= 9.81 \times 0.37$	C1
	$= 3.6 \text{ m s}^{-1}$	A1
2(a)(iii)	$v^2 = 3.6^2 + 4.9^2$	C1
	$v = 6.1 \text{ m s}^{-1}$	A1
	$\theta = \tan^{-1} (3.6 / 4.9)$ $= 36^\circ$	A1
2(b)(i)	$\rho = m / V$	C1
	$V = \frac{4}{3} \pi r^3$	C1
	$\rho = 0.017 / [\frac{4}{3} \pi \times (0.016 / 2)^3]$ $= 7900 \text{ kg m}^{-3}$	A1
2(b)(ii)	$(E =) \frac{1}{2}mv^2$	C1
	$(E =) \frac{1}{2} \times 0.017 \times 4.9^2 = 0.20 \text{ (J)}$	A1
2(b)(iii)	$k = F / x$ or $k = \text{gradient}$	C1
	e.g. $k = 6.4 / 10 \times 10^{-2}$ $= 64 \text{ N m}^{-1}$ (allow 63–65 N m ⁻¹)	A1
2(b)(iv)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
	$x_0 = [(2 \times 0.20) / 64]^{0.5}$ $= 0.079 \text{ m or } 0.080 \text{ m}$	A1
2(c)(i)	same elastic potential energy / same (initial) kinetic energy and (polystyrene ball has) smaller mass (so greater speed) or same (average) force and (polystyrene ball has) smaller mass, (so greater average acceleration so greater speed)	B1
2(c)(ii)	(for the polystyrene ball there is) less (average vertical) acceleration / smaller (average vertical component of) <u>resultant</u> force (so takes longer time to reach ground)	B1

28. M/J 22/P21/Q4

Answer from the official mark scheme: printed pages 328–329 of the scanned booklet are missing

4(a)	$k = F / \Delta L$ or F / x or gradient	C1
	$= \text{e.g. } 30 / (0.60 - 0.20)$	A1
	$= 75 \text{ N m}^{-1}$	
4(b)	$E = \frac{1}{2}F\Delta L$ or $\frac{1}{2}Fx$ or $\frac{1}{2}k(\Delta L)^2$ or $\frac{1}{2}kx^2$ or area under graph	C1
	$= \frac{1}{2} \times 15 \times 0.20$ or $\frac{1}{2} \times 75 \times 0.20^2$	C1
	$= 1.5 \text{ J}$	A1

29. O/N 21/P21/Q3

Answer from the official mark scheme: printed pages 328–329 of the scanned booklet are missing

3(a)(i)	$\sigma = F / xy$	B1
3(a)(ii)	$\varepsilon = (z - w) / w$	B1
3(a)(iii)	$E = \sigma / \varepsilon$	C1
	$= Fw / xy(z - w)$	A1
3(b)(i)	extension = 2.2 mm (<i>allow</i> 2.0–2.4 mm)	A1
3(b)(ii)	strain energy = area under graph/line or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$	C1
	$= \frac{1}{2} \times 120 \times 1.4 \times 10^{-3}$ or $\frac{1}{2} \times 8.6 \times 10^4 \times (1.4 \times 10^{-3})^2$	C1
	$= 0.084 \text{ J}$	A1
3(b)(iii)	(some of the) deformation of the wire is plastic/permanent/not elastic or wire goes past the elastic limit/enters plastic region	B1
	energy (that cannot be recovered) is dissipated as thermal energy/becomes internal energy	B1

30. O/N 21/P23/Q1

Answer from the official mark scheme: printed pages 328–329 of the scanned booklet are missing

1(a)	$m = \rho V$ or ρAL	C1
	$W = mg$	C1
	$(A =) 24 / (9.81 \times 850 \times 0.18) = 0.016 \text{ (m}^2\text{)}$	A1
	or	
	$P = F / A$	(C1)
	$P = \rho gh$	(C1)
	$(A =) 24 / (9.81 \times 850 \times 0.18) = 0.016 \text{ (m}^2\text{)}$	(A1)
1(b)(i)	(upthrust =) $24 + 8(.0) = 32 \text{ (N)}$	A1
1(b)(ii)	$(\Delta)\rho = 32 / 0.016 (= 2000)$	C1
	$(\Delta)\rho = \rho g(\Delta)h$	C1
	$\rho = 2000 / (9.81 \times 0.17)$	
	$= 1200 \text{ kg m}^{-3}$	A1
1(c)(i)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$	C1
	$(\Delta)E = (\frac{1}{2} \times 8.0 \times 0.40) - (\frac{1}{2} \times 4.0 \times 0.20)$ or $(\frac{1}{2} \times 20 \times 0.40^2) - (\frac{1}{2} \times 20 \times 0.20^2)$ or $\frac{1}{2} \times (4.0 + 8.0) \times 0.20$	C1
	$= 1.2 \text{ J}$	A1
1(c)(ii)	length = 30 cm	A1

31. M/J 21/P22/Q1

Answer from the official mark scheme: printed pages 328–329 of the scanned booklet are missing

1(a)	acceleration: vector electrical resistance: scalar momentum: vector <i>1 mark for two correct, 2 marks for all three correct</i>	B2
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1(b)	resultant force (in any direction) is zero	B1
	resultant torque/moment (about any point) is zero	B1
1(c)(i)	upthrust = $\rho g(\Delta)h \times A$	C1
	= $(1.00 \times 10^3 \times 9.81 \times 0.190) \times 0.0230$	A1
	= 42.9 N	
1(c)(ii)	$(T =) 43 - 28 = 15 \text{ (N)}$ or $(T =) 42.9 - 28 = 14.9 \text{ or } 15 \text{ (N)}$	A1
1(c)(iii)	$\sigma = F/A$ or T/A	C1
	= $15 / (3.2 \times 10^{-6})$	A1
	= $4.7 \times 10^6 \text{ Pa}$	
1(c)(iv)	upthrust (on cylinder) increases (and weight constant)	B1
	tension/stress increases and (so) strain energy increases	B1

32. M/J 21/P23/Q3

- (a) force $\times$ distance [2]
perpendicular distance of (line of action of) force from the point
- (b) (i) distance moved by pointer = $123 - 86 (= 37 \text{ mm})$
(extension =) $37 \times (1.8 / 52.6) = 1.3 \text{ (mm)}$
or
 $\sin$ or $\tan \theta = 37 / 526$ (so $\theta = 4.0^\circ$ so extension =) $\sin$ or $\tan \theta \times 18 = 1.3 \text{ (mm)}$ [2]
- (ii) moment = $0.472 \times 9.81 \times 6.2 \times 10^{-2}$ [2]
= 0.29 N m
- (iii) $(\Delta)F \times 1.8 \times 10^{-2} = 0.29$ [2]
 $\Delta F = 16 \text{ N}$
- (iv) $k = F/x$ [2]
= $16 / (1.3 \times 10^{-3})$
= $1.2 \times 10^4 \text{ N m}^{-1}$

33. MAR 21/P22/Q3

- (a) Hooke's (law) [1]
- (b) (i) $k = F/x$ or $k = \text{gradient}$
e.g. $k = 7.0 / 5.0 \times 10^{-2}$ [2]
= 140 N m^{-1}
- (ii) $E = \frac{1}{2} F x$ or $E = \frac{1}{2} k x^2$ or $E = \text{area under graph}$
= $\frac{1}{2} \times 5.6 \times 4.0 \times 10^{-2}$ or $\frac{1}{2} \times 140 \times (4.0 \times 10^{-2})^2$ [2]
= 0.11 J [1]
- (c) (i) (upthrust =) $6.20 - 5.60 = 0.60 \text{ (N)}$
- (ii) $\Delta p = \Delta F / A$
= $0.60 / 1.2 \times 10^{-3}$ [2]
= 500 Pa
- (iii) $(\Delta)p = \rho g(\Delta)h$
 $\rho = 500 / (9.81 \times 5.8 \times 10^{-2})$ [2]
= 880 kg m^{-3} [1]
- (d) (i) (upthrust) increases [1]
(ii) (extension) decreases [1]

34. O/N 20/P22/Q3

(a) (i) $F = kx$
 $F_1 = 800 \times 0.045$
 $= 36 \text{ N}$ [2]

(ii) $(E =) \frac{1}{2}kx^2$ or $\frac{1}{2}Fx$ or area under graph
 $\frac{1}{2} \times 800 \times (0.045)^2$ or $\frac{1}{2} \times 36 \times 0.045 = 0.81 \text{ (J)}$ [2]

(b) (i) efficiency = $(0.72 / 0.81) \times 100$
 $= 89\%$

(ii) $E = \frac{1}{2}mv^2$ [1]
 $p = mv$

$0.72 = \frac{1}{2} \times 0.020 \times v^2$ and $p = 0.020 \times v$
 $p = 0.17 \text{ N s}$

(c) (i) $(\Delta)E = mg(\Delta)h$ [3]

$h = 0.60 / (0.020 \times 9.81) = 3.1 \text{ m}$ [2]

(ii) $F = (0.72 - 0.60) / 3.1$
 $= 0.039 \text{ N}$ [2]

(iii) resultant force on ball is less (than that with air resistance) so time (taken) is more (than T) [1]

35. O/N 20/P21/Q4

(a) (i) (stress =) force / cross-sectional area [1]

(ii) (strain =) extension / original length [1]

(b) (i) $E = FL / Ax$
 $= GL / A$ [2]

(ii) straight line from origin above the original line
line ends at point (4 small squares, F_1). [2]

(iii) 1. shaded area below the graph line and between the two vertical dashed lines [1]

2. remove the force/ F/F_2 and the wire goes back to original length/zero extension [1]

(iv) values have a large range [1]

36. O/N 20/P23/Q4

- (a) compression/extension is proportional to force (provided limit of proportionality is not exceeded) [1]
- (b) $(E) = \frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ or area under graph
 $= \frac{1}{2} \times 8 \times 16 \times 10^{-2} = 0.64 \text{ (J)}$
 or
 $= \frac{1}{2} \times 50 \times (16 \times 10^{-2})^2 = 0.64 \text{ (J)}$ [2]
- (c) (i) $(E) = \frac{1}{2}mv^2$
 $0.64 = \frac{1}{2} \times 0.076 \times v^2$
 $v = 4.1 \text{ m s}^{-1}$ [2]
- (ii) $(\Delta)(E) = mg(\Delta)h$
 $= 0.076 \times 9.81 \times 0.24$
 $(= 0.18 \text{ (J)})$
 kinetic energy = $0.64 - 0.18$
 $= 0.46 \text{ J}$ [3]
- (iii) $v = 4.1 \text{ m s}^{-1}$ [1]
- (d) $W = Fs$
 $d = 0.30 + (2\pi \times 0.12) + 0.25 (= 1.3 \text{ m})$
 $F = 0.23 / 1.3$
 $= 0.18 \text{ N}$ [3]

37. M/J 20/P22/Q5

- (a) Hooke's (law) [1]
- (b) (i) $\sigma = F / A$
 $= 36 / (4.1 \times 10^{-7})$
 $= 8.8 \times 10^7 \text{ Pa}$ [2]
- (ii) Young modulus = σ / ϵ or $F / A\epsilon$
 $\epsilon = 8.8 \times 10^7 / (1.7 \times 10^{11})$
 $= 5.2 \times 10^{-4}$ [2]
- (c) $R = \rho L / A$
 $\Delta R = \rho \Delta x / A$
 $= 3.7 \times 10^{-7} \times 0.12 \times 10^{-3} / (4.1 \times 10^{-7})$
 $= 1.1 \times 10^{-4} \Omega$ [3]
- (d) remove the force/ F and wire returns to original length [1]

38. M/J 20/P21/Q3

- (a) resultant force (in any direction) is zero [2]
 resultant torque/moment (about any point) is zero
- (b) (i) 1. $T \sin 53^\circ = 2.4$
 $T = 3.0 \text{ N}$
 2. $F = T \cos 53^\circ$ or $F^2 = T^2 - 2.4^2$
 $F = 1.8 \text{ N}$ [2]
- (ii) $\sigma = T / A$ or $\sigma = F / A$
 $A = \pi d^2 / 4$ or $A = \pi r^2$
 $\sigma = 3.0 \times 4 / [\pi \times (0.50 \times 10^{-3})^2]$
 $= 1.5 \times 10^7 \text{ Pa}$ [3]
 [1]
- (c) (i) $h = 75 - 75 \sin 53^\circ = 15 \text{ cm}$
 (ii) $(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$
 $(\Delta)E = 2.4 \times 15 \times 10^{-2}$
 $= 0.36 \text{ J}$ [2]
- (iii) $E = \frac{1}{2}mv^2$
 $0.36 = \frac{1}{2} \times (2.4 / 9.81) \times v^2$
 $v = 1.7 \text{ m s}^{-1}$ [3]

39. O/N 19/P21/Q4

- (a) $k = F / x$ or $k = \text{gradient}$
 e.g. $k = 4.0 / 0.050$
 $k = 80 \text{ N m}^{-1}$ [2]
- (b) $E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$
 $(\Delta)E = (\frac{1}{2} \times 3.2 \times 0.040) - (\frac{1}{2} \times 1.2 \times 0.015) = 0.055 \text{ J}$
 or
 $(\Delta)E = (\frac{1}{2} \times 80 \times 0.040^2) - (\frac{1}{2} \times 80 \times 0.015^2) = 0.055 \text{ J}$
 or
 $(\Delta)E = \frac{1}{2} \times (1.2 + 3.2) \times 0.025 = 0.055 \text{ J}$ [2]
- (c) $(\Delta)E = mg(\Delta)h$
 $= 0.122 \times 9.81 \times (0.120 - 0.095)$
 $= 0.030 \text{ J}$
 or
 $(\Delta)E = W \times (\Delta)h$
 $= 1.2 \times 0.025$
 $= 0.030 \text{ J}$ [2]
- (d) (i) $E = 0.055 - 0.030$
 $= 0.025 \text{ J}$ [1]
- (ii) $E = \frac{1}{2}mv^2$
 $v = [(2 \times 0.025) / 0.122]^{0.5}$
 $= 0.64 \text{ m s}^{-1}$ [2]

40. O/N 19/P23/Q4/b

- (a) (i) $E = \frac{1}{2}mv^2$
 $\frac{1}{2} \times 0.50 \times 2.4^2 = \frac{1}{2} \times 72 \times x^2$
 $x = 0.20 \text{ m}$ [3]
- (ii) 1. straight line from the origin sloping upwards
 2. line drawn from a positive value of E_k at $x = 0$ to a positive value of x at $E_k = 0$
 line has an increasing downwards slope [3]

41. M/J 19/P22/Q2/c,d

(a) $E = \sigma / \epsilon$

$E = (F/A) / (\Delta L/L) \text{ or } E = FL/A\Delta L$

$$\Delta L = (480 \times 0.48) / (3.0 \times 10^{-4} \times 2.2 \times 10^{11})$$
$$= 3.5 \times 10^{-6} \text{ m}$$

[3]

(b) $f_o = f_s v / (v - v_s)$

$480 = f_s \times 340 / (340 - 14)$

$f_s = 460 \text{ Hz}$

[2]

42. M/J 19/P21/Q3

(a) (i) $E = \frac{1}{2}Fx \text{ or } E = \frac{1}{2}kx^2 \text{ or } E = \text{area under graph}$

$E = \frac{1}{2} \times 4.0 \times 0.32 = 0.64 \text{ J} \text{ or } E = \frac{1}{2} \times 12.5 \times (0.32)^2 = 0.64 \text{ J}$

[2]

(ii) $E = mgh \text{ or } E = Wh$

$= 2.5 \times 0.32$

$= 0.80 \text{ J}$

[2]

(b) (i) kinetic energy = $0.80 - 0.64$

$= 0.16 \text{ J}$

[1]

(ii) $E = \frac{1}{2}mv^2$

$0.16 = \frac{1}{2} \times (2.5 / 9.81) \times v^2$

$v = 1.1 \text{ m s}^{-1}$

[2]

43. O/N 18/P23/Q1**(a)** current

temperature

(allow amount of substance, luminous intensity)*any two correct answers, 1 mark each*

[2]

(b) (i) $W = 2 \times (150 \times \sin 17^\circ) \text{ or } 2 \times (150 \times \cos 73^\circ)$

$W = 88 \text{ N}$

[2]

(ii) 1. $\sigma = F/A$

$= 150 / (7.5 \times 10^{-6})$

$= 2.0 \times 10^7 \text{ Pa}$

[2]

2. $\epsilon = \sigma / E$

$= 2.0 \times 10^7 / (2.1 \times 10^{11})$

$= 9.5 \times 10^{-5}$

[2]

44. M/J 18/P21/Q2

Answer from the official mark scheme

2(a)	a body continues at (rest or) constant velocity unless acted upon by a resultant force	B1
2(b)(i)1.	from 0–2 s, distance = $\frac{1}{2} \times 2 \times 6.8 (= 6.8 \text{ m})$	C1
	and from 2–3 s, distance = $\frac{1}{2} \times 1 \times 3.4 (= 1.7 \text{ m})$	
	magnitude of displacement = 5.1 m	A1
	direction of displacement is down(wards)	B1
2(b)(i)2.	$(\Delta E) = mg\Delta h \text{ or } (E) = mgh \text{ or } (E) = Wh$	C1
	$(\Delta)E = 15 \times 5.1$	A1
	= (–) 77 J	
2(b)(ii)	$a = (v - u) / t \text{ or } a = \text{gradient} \text{ or } a = dv/dt$	C1
	$a = 3.4 \text{ ms}^{-2}$	A1

2(b)(iii)	$T - W = ma$ or $T - mg = ma$	C1
	$T = 15 + (15/9.81) \times 3.4 = 20\text{ N or } 20.2\text{ N}$	A1
2(b)(iv)	$E = F/A\epsilon$ or $E = \sigma/\epsilon$ and $\sigma = F/A$	C1
	$\epsilon = 20 / (2.8 \times 10^{-5} \times 1.7 \times 10^{11})$	C1
	$= 4.2 \times 10^{-6}$	A1
2(b)(v)	block is in equilibrium/has no resultant force	B1
	block could be stationary (or have constant velocity/speed) (so no, not possible to deduce)	B1

45. M/J 18/P23/Q4

(a) (Young modulus =) stress / strain [1]

(b) (i) $k = F/\Delta L$ or 1/gradient
 $= 90 \times 10^3 / (2 \times 10^{-3})$ (or other point on line)
 $= 4.5 \times 10^7 \text{ Nm}^{-1}$ [2]

(ii) $E = \frac{1}{2}F\Delta L$ or $E = \frac{1}{2}k(\Delta L)^2$
 $= \frac{1}{2} \times 90 \times 10^3 \times 2 \times 10^{-3}$ or $\frac{1}{2} \times 4.5 \times 10^7 \times (2 \times 10^{-3})^2$
 $= 90\text{ J}$ [3]

(c) straight line starting from (0, 150) and below original line
line ends at (90, 147) [2]

46. MAR 18/P22/Q3

(a) (i) force / (cross-sectional) area [1]
(ii) extension / original length [1]

(b) (i) measure / determine / find diameter
using a micrometer / digital calipers
several measurements in different places / along the wire / around the circumference
(and average them) [3]

(ii) $E = \sigma/\epsilon$ or $E = FL/Ax$ or $E = \text{gradient} \times (L/A)$
 $E = (4 \times 2.5) / (0.8 \times 10^{-3}) \times (9.4 \times 10^{-8})$
 $= 1.3 \times 10^{11} \text{ Pa}$ [2]

(iii) $E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$
 $E = \frac{1}{2} \times (2+4) \times 0.4 \times 10^{-3}$
or $E = (\frac{1}{2} \times 4 \times 0.8 \times 10^{-3}) - (\frac{1}{2} \times 2 \times 0.4 \times 10^{-3})$
or $E = [\frac{1}{2} \times 5000 \times (0.8 \times 10^{-3})^2] - [\frac{1}{2} \times 5000 \times (0.4 \times 10^{-3})^2]$
 $E = 1.2 \times 10^{-3} \text{ J}$

(c) straight line from the origin and above the original line
straight line passes through (0.80, 8.0) [2]

47. O/N 17/P22/Q3

- (a) $p = 1000 \times 9.81 \times 7.0 \times 10^{-2}$ or $1000 \times 9.81 \times 1.9 \times 10^{-2}$
 $\Delta p = 1000 \times 9.81 \times (7.0 \times 10^{-2} - 1.9 \times 10^{-2})$ or $686 - 186$
 $= 500 \text{ Pa}$ [2]
- (b) $F = pA$ or $(\Delta)F = \Delta p \times A$
upthrust $= 500 \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$
or
upthrust $= (686 - 186) \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$
or
upthrust $= 1000 \times 9.81 \times 5.1 \times 10^{-2} \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$ [2]
- (c) force $= 4.0 - 1.3$
 $= 2.7 \text{ N}$ [1]
- (d) extension/ x/e $= 2.7/30$
 $= 0.09 \text{ (m)}$ or 9 (cm)
height above surface $= 9 - 7$
 $= 2 \text{ cm}$ [3]
- (e) (i) mass $= 4.0/9.81$
acceleration $= 2.7/(4.0/9.81)$
 $= 6.6 \text{ m s}^{-2}$ [2]
- (ii) viscous force increases (and then becomes constant)
(weight and upthrust constant so) acceleration decreases (to zero) [2]

48. O/N 17/P21/Q4

- (a) (strain =) extension / original length [1]
- (b) (i) $E = \sigma/\epsilon$
maximum stress $= 2.1 \times 10^{11} \times 4.0 \times 10^{-4}$
 $= 8.4 \times 10^7 \text{ Pa}$ [2]
- (ii) $\sigma = F/A$
minimum area $= 8.0 \times 10^3/8.4 \times 10^7$
 $= 9.5 \times 10^{-5} \text{ m}^2$ [2]

49. M/J 17/P22/Q3

- (a) $E = \text{stress} / \text{strain}$ or $(F/A) / (e/l)$
 $= [\text{gradient} \times 3.5] / [\pi \times (0.19 \times 10^{-3})^2]$
e.g. $E = \{[(40 - 5) / ([11.6 - 3.2] \times 10^{-3})] \times 3.5\} / [\pi \times (0.19 \times 10^{-3})^2]$
or
 $[4170 \times 3.5] / [\pi \times (0.19 \times 10^{-3})^2]$
 $E (= 1.3 \times 10^{11}) = 0.13 \text{ TPa}$ (allow answers in range 0.120–0.136 TPa) [3]
- (b) a larger range of F required or range greater than 35 N [1]

50. M/J 17/P23/Q4

- (a) the straight line does not go through the origin/the force is not proportional to extension (so does not obey Hooke's law) [1]
- (b) elastic potential energy [1]
- (c) remove the force/masses and the spring returns to its original length if elastic [1]
- (d) work done is represented by/linked to area under the line ($\times g$)
work $= \frac{1}{2} (145 + 70) \times 10^{-3} \times 9.81 \times 120 \times 10^{-3}$
 $= 0.13 \text{ (0.127) J}$ [3]

51. O/N 16/P21/Q3

- (a) force/load is proportional to extension/compression (provided proportionality limit is not exceeded) [1]
- (b) (i) $k = F/x$ or $k = \text{gradient}$
 $k = 600 \text{ N m}^{-1}$ [2]
- (ii) $(W =) \frac{1}{2}kx^2$ or $(W =) \frac{1}{2}Fx$ or $(W =) \text{area under graph}$
 $(W =) 0.5 \times 600 \times (0.040)^2 = 0.48 \text{ J}$ or $(W =) 0.5 \times 24 \times 0.040 = 0.48 \text{ J}$ [2]
- (iii) 1. $(E_K =) \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 0.025 \times 6.0^2$
 $= 0.45 \text{ J}$ [2]
2. (work done against resistive force $=$) $0.48 - 0.45 [= 0.03(0) \text{ J}]$
average resistive force $= 0.030/0.040$
 $= 0.75 \text{ N}$ [3]
- (iv) efficiency $= [\text{useful energy out}/\text{total energy in}] (\times 100)$
 $= [0.45/0.48] (\times 100)$
 $= 0.94$ or 94% [2]

52. M/J 16/P22/Q3

- (a) Young modulus $= \text{stress}/\text{strain}$ [1]
- (b) (i) $E = (F \times l)/(A \times e)$ or $e = (F \times l)/(A \times E)$
 $e \propto 1/E$
or
ratio $e_C/e_S = E_S/E_C$ or $(1.9 \times 10^{11})/(1.2 \times 10^{11})$ or $19/12$
(ratio $=$) 1.6 (1.58) [3]
- (ii) two straight lines from $(0,0)$ with **S** having the steepest gradient [1]

53. M/J 16/P21/Q4

- (a) the energy (stored) in a body due to its extension/compression/deformation/change in shape/size [1]
- (b) (i) two values of F/x are calculated which are the same
e.g. $10.4/40 = 0.26$ and $6.5/25 = 0.26$
or
ratio of two forces and the ratio of the corresponding two extensions are calculated which are the same
e.g. $5.2/10.4 = 0.5$ and $20/40 = 0.5$
or
gradient of graph line calculated and coordinates of one point on the line used with straight line equation $y = mx + c$ to show $c = 0$
(so) force is proportional to extension (and so Hooke's law obeyed) [2]
- (ii) 1. $k = F/x$ or $k = \text{gradient}$
gradient or values from a single point used e.g. $k = 10.4/(40 \times 10^{-2})$
 $k = 26 \text{ N m}^{-1}$ [2]
2. work done $=$ area under graph
or $\frac{1}{2}Fx$ or $\frac{1}{2}(F_2 + F_1)(x_2 - x_1)$
or $\frac{1}{2}kx^2$ or $\frac{1}{2}k(x_2^2 - x_1^2)$
 $= \frac{1}{2} \times 10.4 \times 0.4 - \frac{1}{2} \times 5.2 \times 0.2$
or $\frac{1}{2} \times (5.2 + 10.4) \times 20 \times 10^{-2}$
or $\frac{1}{2} \times 26 \times (0.4^2 - 0.2^2)$
 $= 1.6 \text{ J}$ [3]
- (c) remove the force and the spring goes back to its original length [1]

54. O/N 15/P22/Q3

- (a) (i) $k = F/x$ or $1/\text{gradient}$
 $(k = 4.4/(5.4 \times 10^{-2}) =) 81 \text{ (81.48) N m}^{-1}$ [2]
- (ii) work done = area under line or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$
 $(= 0.5 \times 4.4 \times 5.4 \times 10^{-2} =) 0.12 \text{ (0.119) J}$ [2]
- (b) (i) kinetic energy/ E_k of trolley/T (and block) changes to EPE/strain energy/elastic energy of spring
 EPE changes to KE of trolley/T and KE of block or to give lower KE to trolley [2]
- (ii) change in momentum = $m(v + u)$
 $= 0.25 (0.75 + 1.2) = 0.49 \text{ (0.488) N s}$ [2]

55. O/N 15/P23/Q7

- (a) stress or $\sigma = F/A$
 max. tension = $UTS \times A = 4.5 \times 10^8 \times 15 \times 10^{-6} = 6800 \text{ (6750) N}$ [2]
- (b) $\rho = m/V$
 weight = $mg = \rho Vg = \rho ALg$
 $6750 = 7.8 \times 10^3 \times 15 \times 10^{-6} \times L \times 9.81$
 $L = 5.9 \text{ (5.88) } \times 10^3 \text{ m}$
 or
 maximum mass = $6750/9.81 = 688 \text{ kg}$
 mass per unit length = $\rho A = 0.117 \text{ kg m}^{-1}$
 $L = 688/0.117 = 5.9 \times 10^3 \text{ m}$
 or
 maximum mass = $6750/9.81 = 688 \text{ kg}$
 volume = $m/\rho = 0.0882 \text{ m}^3 = LA$
 $L = 0.0882/15 \times 10^{-6} = 5.9 \times 10^3 \text{ m}$ [3]